AWS Networking Workshop
- Foundational Topics
  - Prerequisites
  - VPC Fundamentals
  - Multiple VPCs
    - VPC Peering
    - Transit Gateway
    - TGW Route Tables
    - Clean Up
  - Security Controls
  - Connecting to On-Premises
  - Connecting to SDWAN
  - Network Monitoring
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Workshop catalog in AWS Builder Center

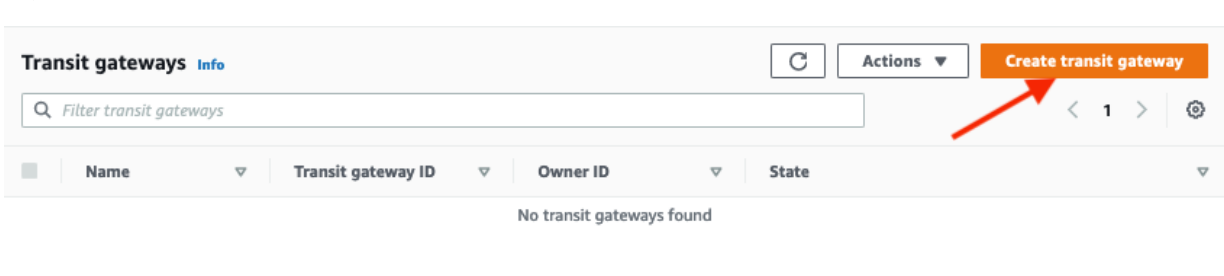
Content preferences
Language
English

Transit Gateway

AWS Transit Gateway connects your Amazon Virtual Private Clouds (VPCs) and on-premises networks through a central hub. This simplifies your network and puts an end to complex peering relationships. It acts as a cloud router – each new connection is only made once.

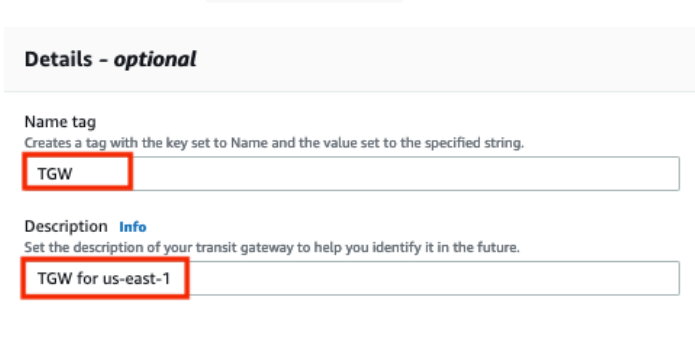
Create the Transit Gateway

1. In the left hand pane of the VPC Dashboard scroll down and click on Transit Gateways
2. Click on **Create Transit Gateway**



Transit gateways Info
Actions
Create transit gateway
Filter transit gateways
Name
Transit gateway ID
Owner ID
State
No transit gateways found

3. Add a name for the new Transit Gateway as `TGW` and add a description of `TGW for us-east-1`.

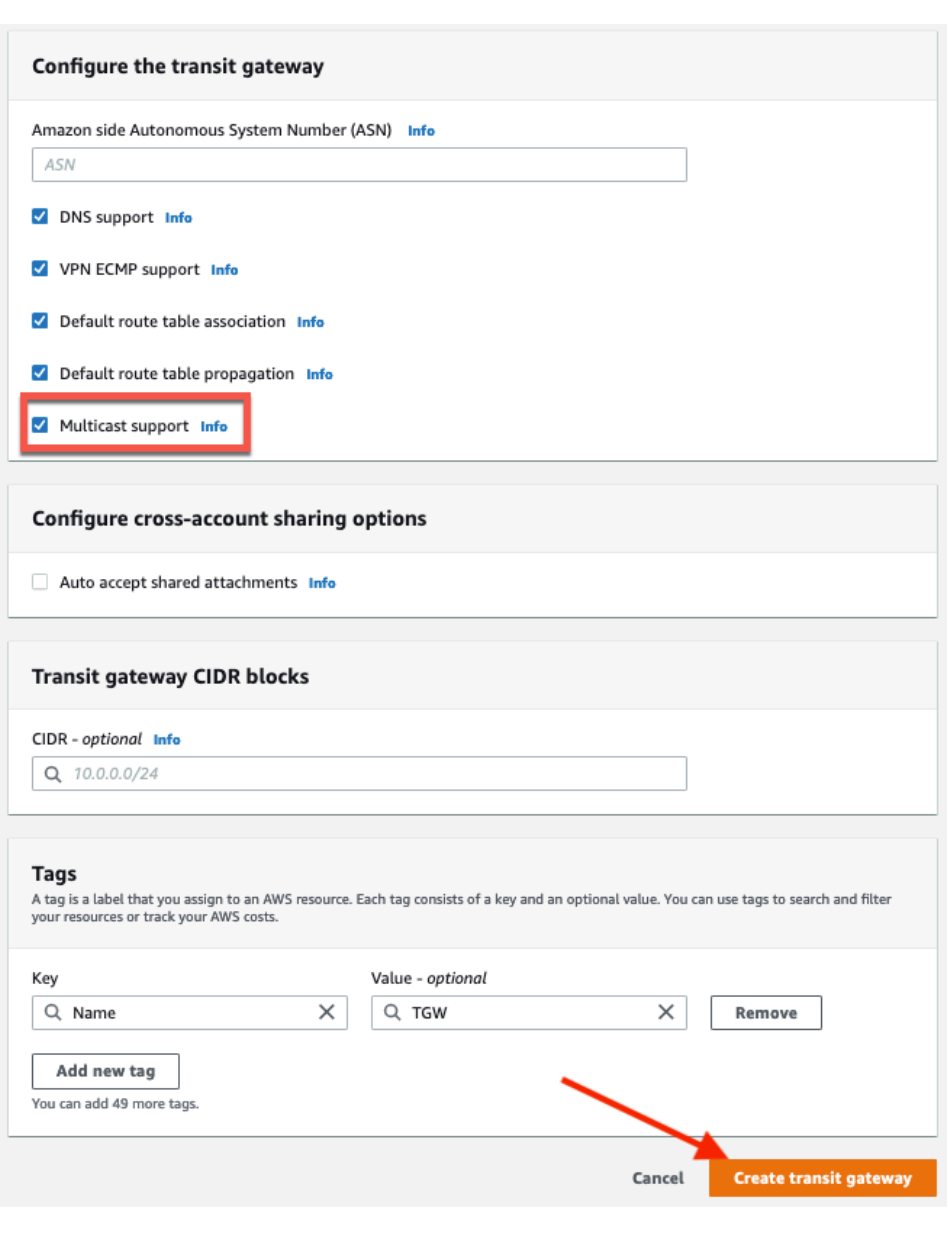


Details - *optional*

Name tag: TGW

Description: TGW for us-east-1

4. Select `Multicast support` and keep the remaining settings at the defaults. You will need this option enabled if you progress to the advanced multicast lab. Click on **Create transit gateway**.



Configure the transit gateway

Amazon side Autonomous System Number (ASN): ASN

☑ DNS support

☑ VPN ECMP support

☑ Default route table association

☑ Default route table propagation

☑ Multicast support

Configure cross-account sharing options

☐ Auto accept shared attachments

Transit gateway CIDR blocks

CIDR - *optional*: 10.0.0.0/24

Tags

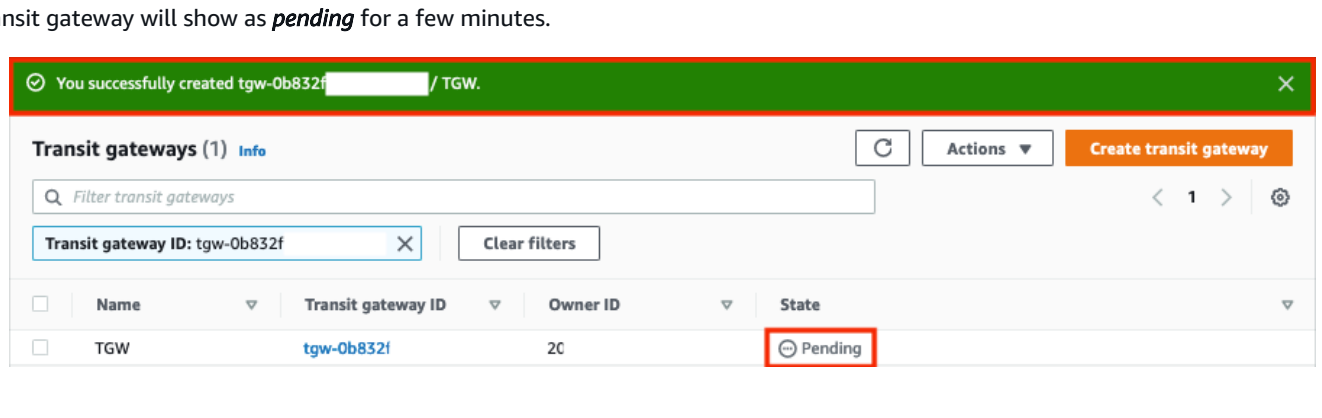
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key: Name Value - *optional*: TGW

You can add 49 more tags.

Remove
Add new tag
Cancel
Create transit gateway

5. The state of the new transit gateway will show as ***pending*** for a few minutes.



You successfully created tgw-0b832f[redacted]/ TGW.

Transit gateways (1)

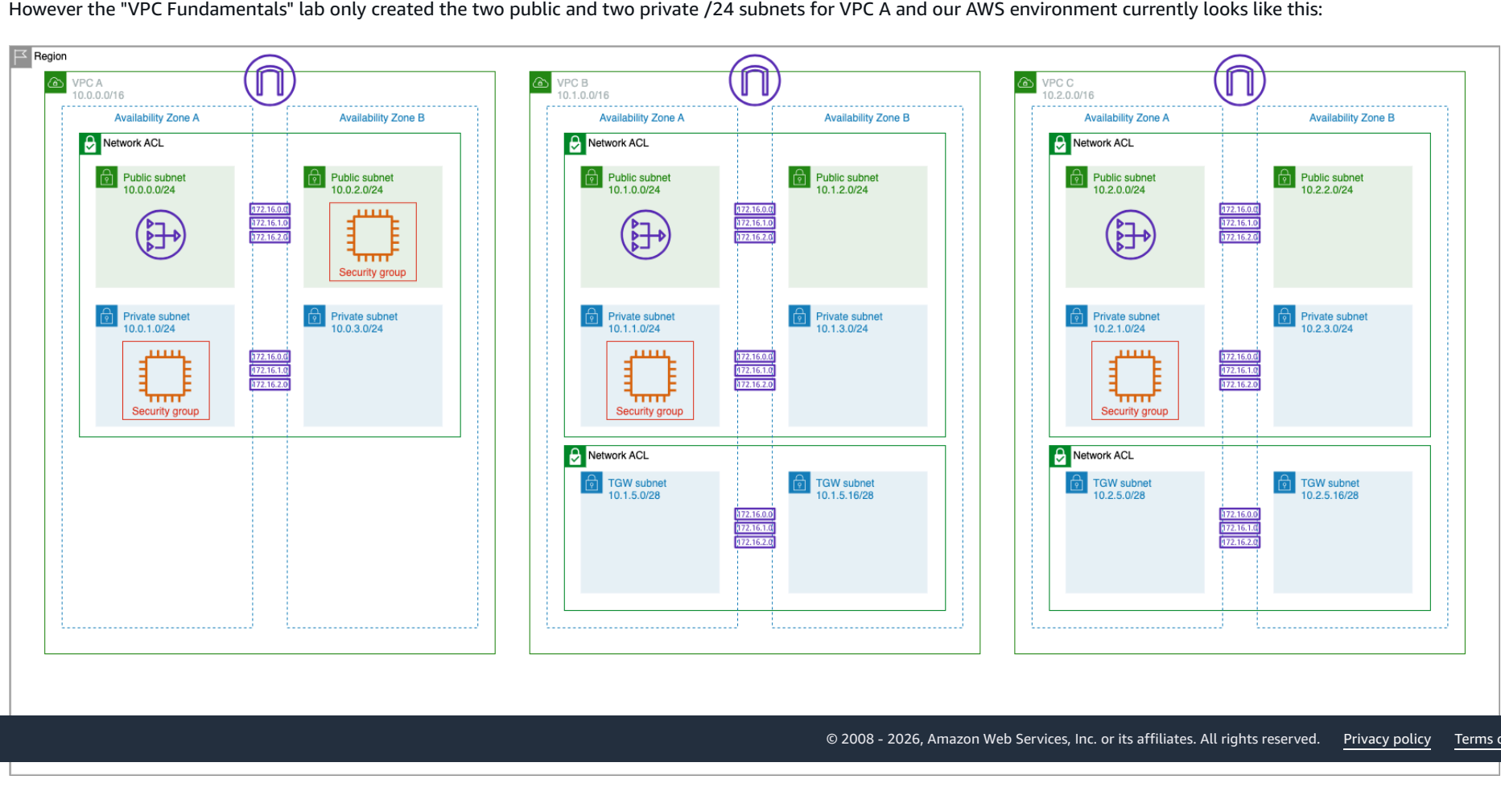
Transit gateway ID: tgw-0b832f

Name	Transit gateway ID	Owner ID	State
TGW	tgw-0b832f	2C	Pending

Attach VPCs to Transit Gateway

The best practice for connecting VPCs to Transit Gateway is to use a dedicated /28 subnet in each availability zone and the CloudFormation run earlier created these for VPC B and VPC C alongside two private and public /24 subnets for hosting workloads.

However the "VPC Fundamentals" lab only created the two public and two private /24 subnets for VPC A and our AWS environment currently looks like this:

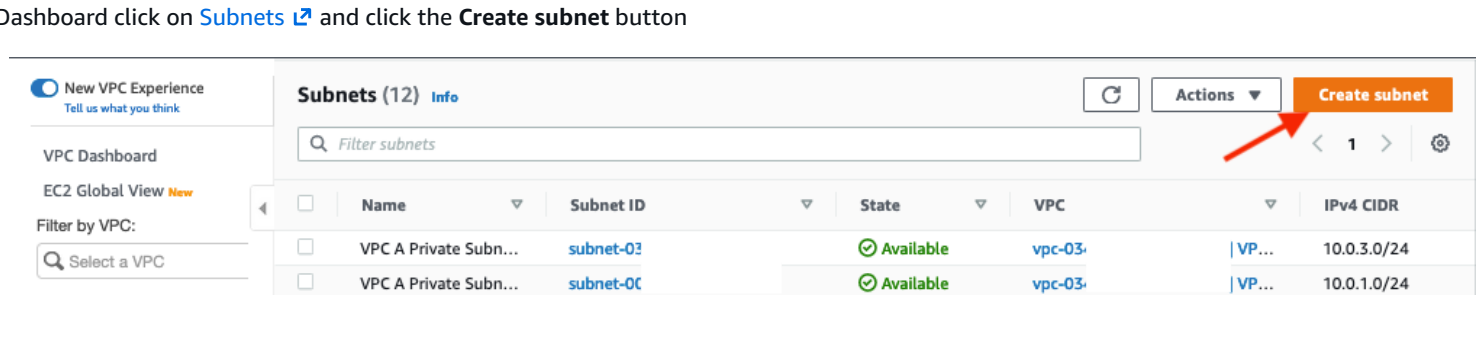


Region
VPC A 10.0.0.0/16
VPC B 10.1.0.0/16
VPC C 10.2.0.0/16
Availability Zone A
Availability Zone B
Network ACL
Public subnet 10.0.0.0/24
Public subnet 10.0.2.0/24
Private subnet 10.0.1.0/24
Private subnet 10.0.3.0/24
Public subnet 10.1.0.0/24
Public subnet 10.1.2.0/24
Private subnet 10.1.1.0/24
Private subnet 10.1.3.0/24
TGW subnet 10.1.5.0/28
TGW subnet 10.1.5.16/28
Public subnet 10.2.0.0/24
Public subnet 10.2.2.0/24
Private subnet 10.2.1.0/24
Private subnet 10.2.3.0/24
TGW subnet 10.2.5.0/28
TGW subnet 10.2.5.16/28
Security group

Therefore before we create the transit gateway we need to add a dedicated /28 subnet in each availability zone in **VPC A** for the transit gateway attachments.

Create Transit Gateway Subnets in VPC A

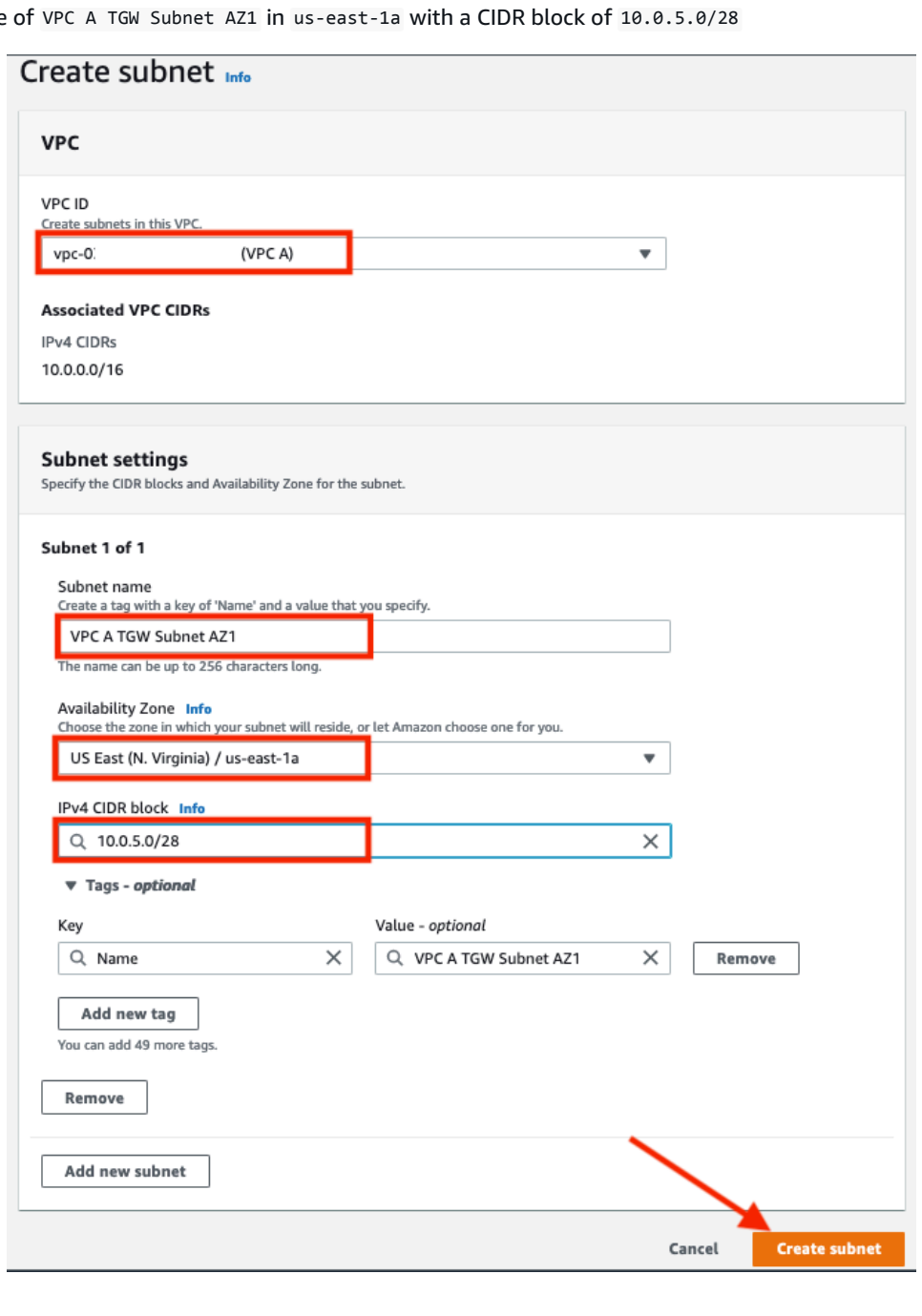
1. Within the VPC Dashboard click on Subnets and click the **Create subnet** button



New VPC Experience
Tell us what you think
VPC Dashboard
EC2 Global View New
Filter by VPC:
Select a VPC
Subnets (12) Info
Actions
Create subnet
Filter subnets

Name	Subnet ID	State	VPC	IPv4 CIDR
VPC A Private Subn…	subnet-03	Available	vpc-03- | VP…	10.0.3.0/24
VPC A Private Subn…	subnet-0C	Available	vpc-03- | VP…	10.0.1.0/24

2. Create a subnet under VPC A with a name of `VPC A TGW Subnet AZ1` in `us-east-1a` with a CIDR block of `10.0.5.0/28`



Create subnet

VPC

VPC ID: vpc-0 (VPC A)

Associated VPC CIDRs

IPv4 CIDRs: 10.0.0.0/16

Subnet settings

Subnet 1 of 1

Subnet name: VPC A TGW Subnet AZ1

Availability Zone: US East (N. Virginia) / us-east-1a

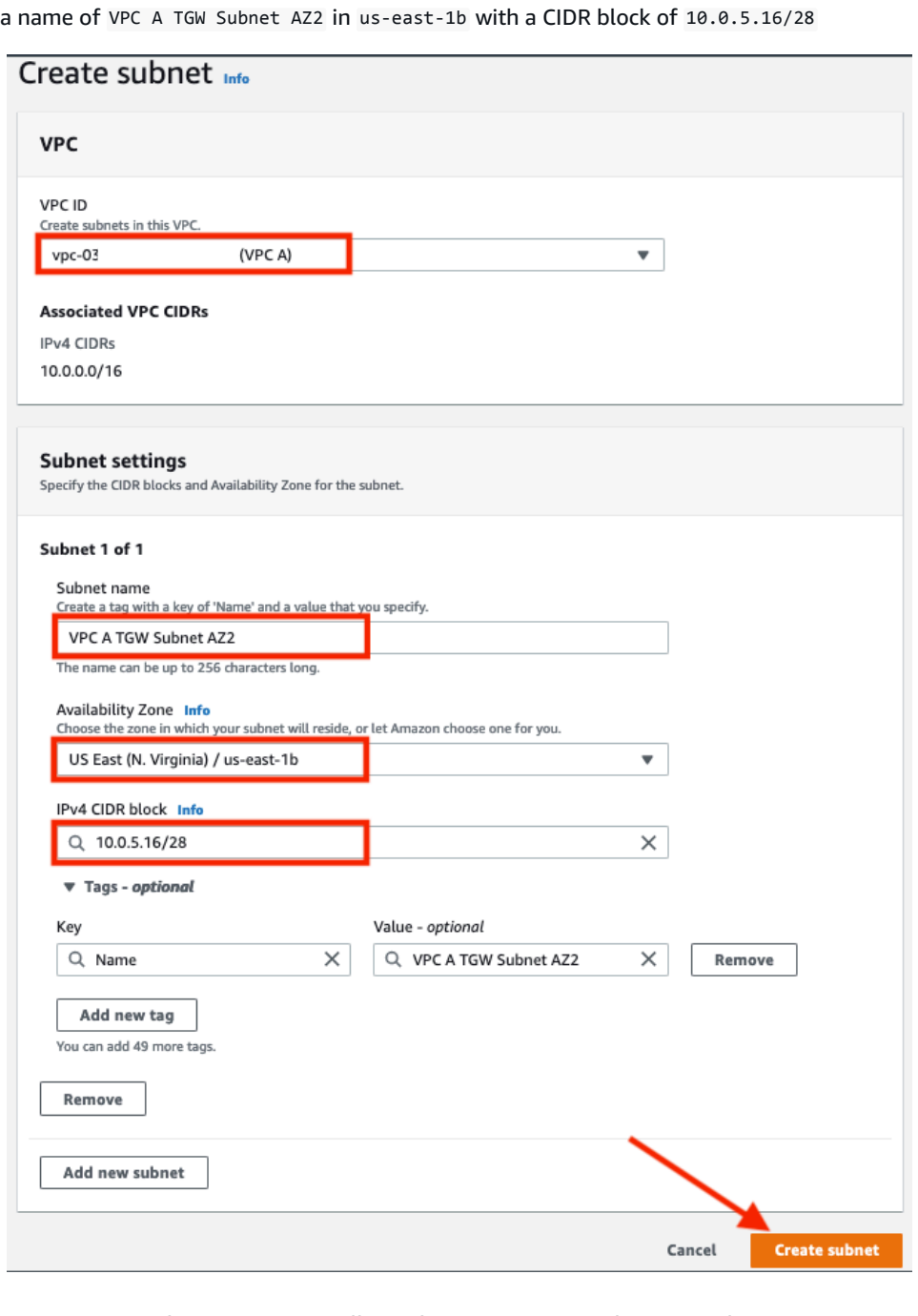
IPv4 CIDR block: 10.0.5.0/28

Tags - *optional*

Key: Name Value - *optional*: VPC A TGW Subnet AZ1

Remove
Add new tag
Add new subnet
Cancel
Create subnet

3. Create another subnet under VPC A with a name of `VPC A TGW Subnet AZ2` in `us-east-1b` with a CIDR block of `10.0.5.16/28`



Create subnet

VPC

VPC ID: vpc-03 (VPC A)

Associated VPC CIDRs

IPv4 CIDRs: 10.0.0.0/16

Subnet settings

Subnet 1 of 1

Subnet name: VPC A TGW Subnet AZ2

Availability Zone: US East (N. Virginia) / us-east-1b

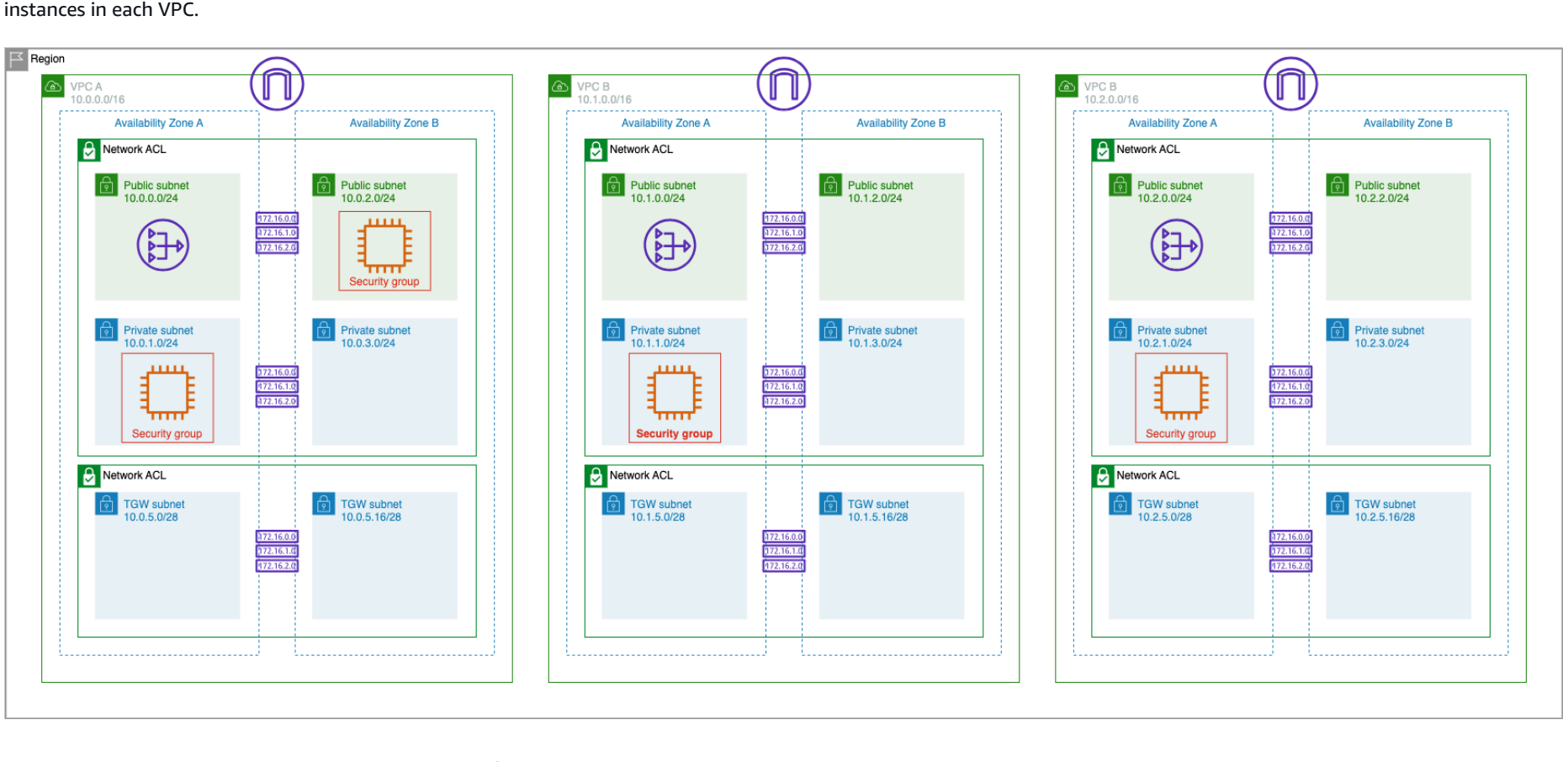
IPv4 CIDR block: 10.0.5.16/28

Tags - *optional*

Key: Name Value - *optional*: VPC A TGW Subnet AZ2

Remove
Add new tag
Add new subnet
Cancel
Create subnet

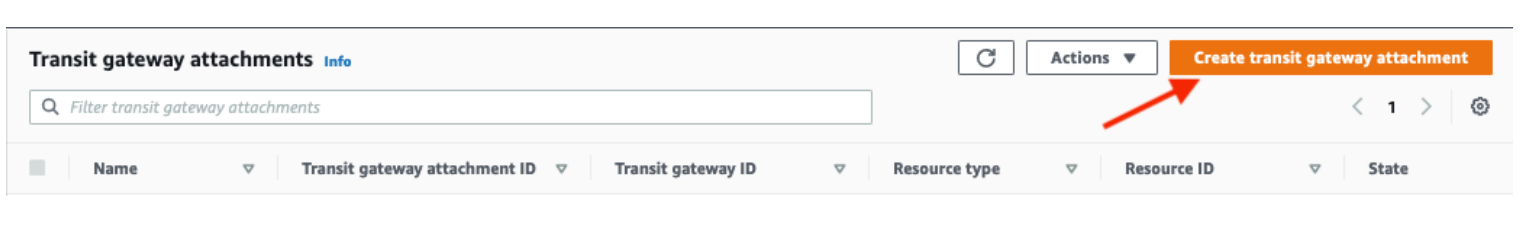
Now that we have subnets to place the transit gateway attachments into we will attach VPC A, VPC B, and VPC C to the transit gateway and test connectivity between our EC2 instances in each VPC.



Region
VPC A 10.0.0.0/16
VPC B 10.1.0.0/16
VPC B 10.2.0.0/16
Availability Zone A
Availability Zone B
Network ACL
Public subnet 10.0.0.0/24
Public subnet 10.0.2.0/24
Private subnet 10.0.1.0/24
Private subnet 10.0.3.0/24
TGW subnet 10.0.5.0/28
TGW subnet 10.0.5.16/28
Public subnet 10.1.0.0/24
Public subnet 10.1.2.0/24
Private subnet 10.1.1.0/24
Private subnet 10.1.3.0/24
TGW subnet 10.1.5.0/28
TGW subnet 10.1.5.16/28
Public subnet 10.2.0.0/24
Public subnet 10.2.2.0/24
Private subnet 10.2.1.0/24
Private subnet 10.2.3.0/24
TGW subnet 10.2.5.0/28
TGW subnet 10.2.5.16/28
Security group

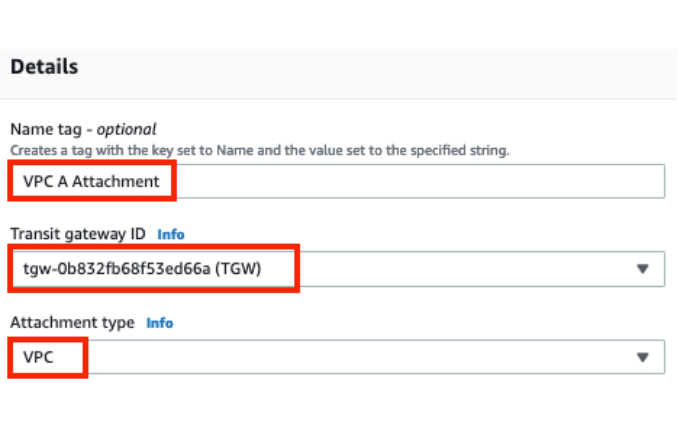
Create Transit Gateway Attachment for VPC A

1. On the left navigation pane go to Transit Gateway Attachments
2. Click on **Create Transit Gateway Attachment**.



Transit gateway attachments Info
Actions
Create transit gateway attachment
Filter transit gateway attachments
Name
Transit gateway attachment ID
Transit gateway ID
Resource type
Resource ID
State

3. Enter `VPC A Attachment` as the **Name tag**.
4. Select the transit gateway from the dropdown for **Transit Gateway ID**.
5. Leave **Attachment Type** as `VPC`



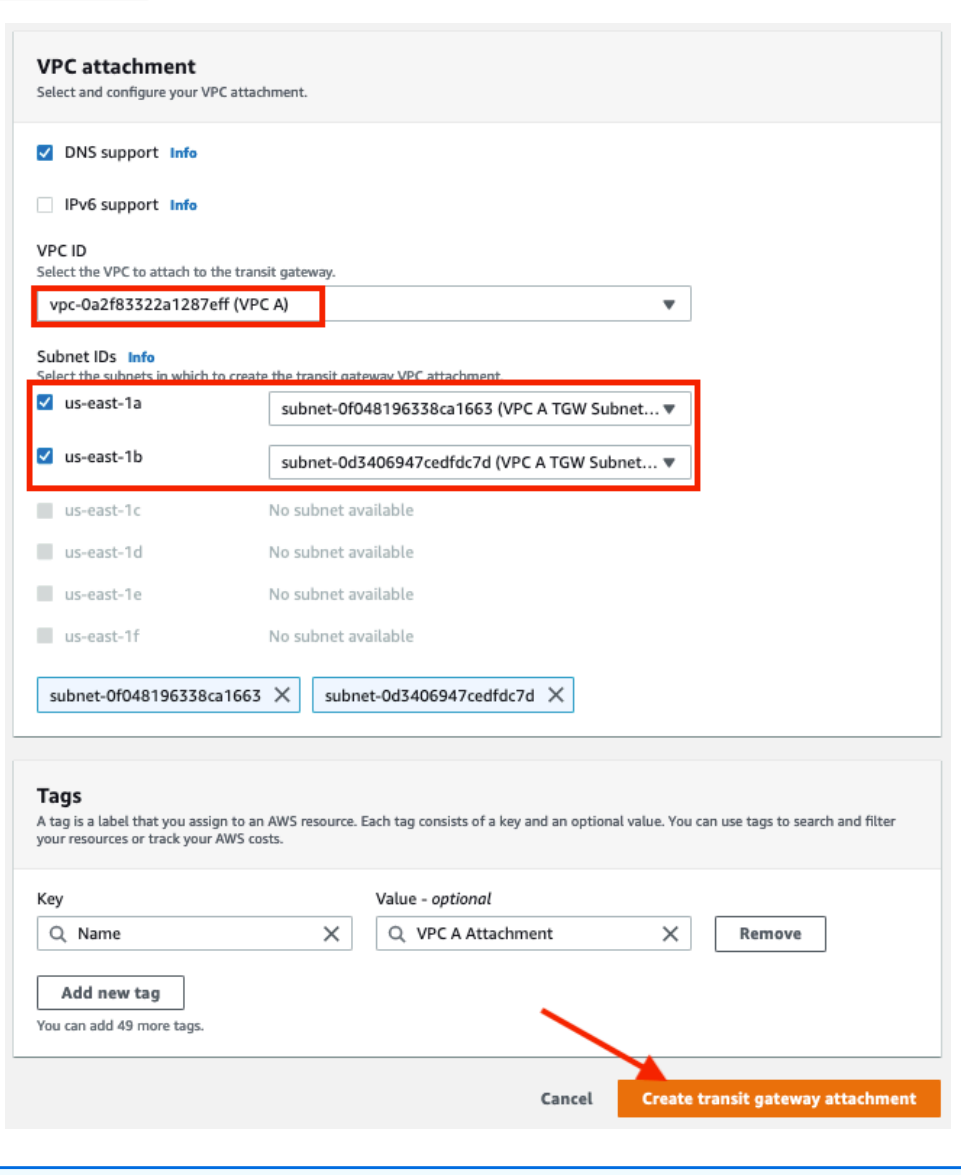
Details

Name tag - *optional*: VPC A Attachment

Transit gateway ID: tgw-0b832fb68f53ed66a (TGW)

Attachment type: VPC

6. Select **VPC A** from the **VPC ID** drop down.
7. Select `VPC A TGW Subnet AZ1` and `VPC A TGW Subnet AZ2` for the **Subnet IDs**.



VPC attachment

Select and configure your VPC attachment.

☑ DNS support

☐ IPv6 support

VPC ID: vpc-0a2f83322a1287eff (VPC A)

Subnet IDs

☑ us-east-1a subnet-0f048196338ca1663 (VPC A TGW Subnet…

☑ us-east-1b subnet-0d3406947cedfdc7d (VPC A TGW Subnet…

☐ us-east-1c No subnet available

☐ us-east-1d No subnet available

☐ us-east-1e No subnet available

☐ us-east-1f No subnet available

subnet-0f048196338ca1663 subnet-0d3406947cedfdc7d

Tags

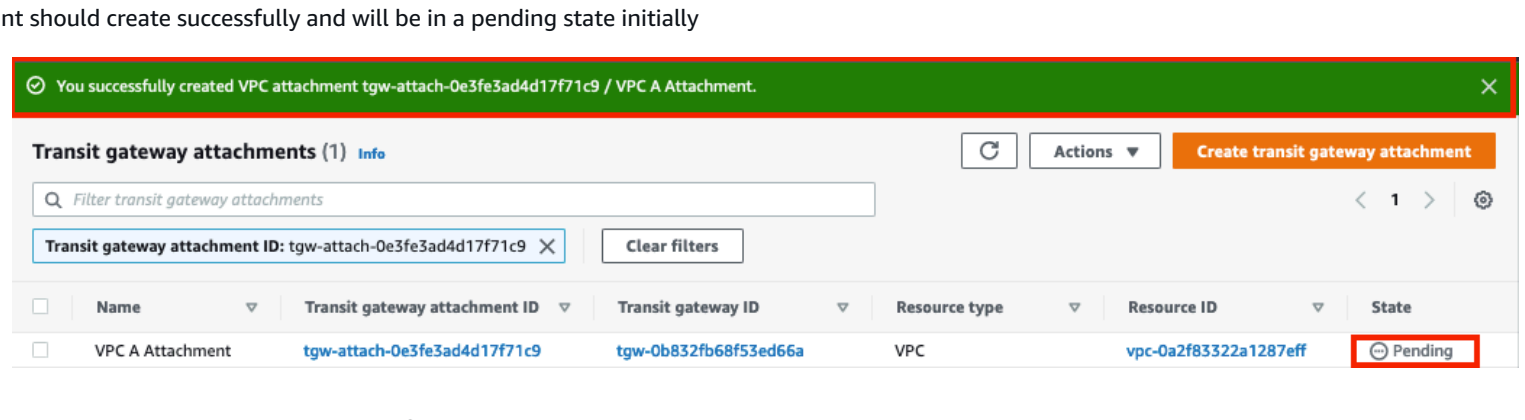
Key: Name Value - *optional*: VPC A Attachment

You can add 49 more tags.

Remove
Add new tag
Cancel
Create transit gateway attachment

> Note that the TGW subnets will not be selected by default, double check the subnets are the TGW ones.

8. Click **Create transit gateway attachment** on the bottom right corner.
9. VPC attachment should create successfully and will be in a pending state initially



You successfully created VPC attachment tgw-attach-0e3fe3ad4d17f71c9 / VPC A Attachment.

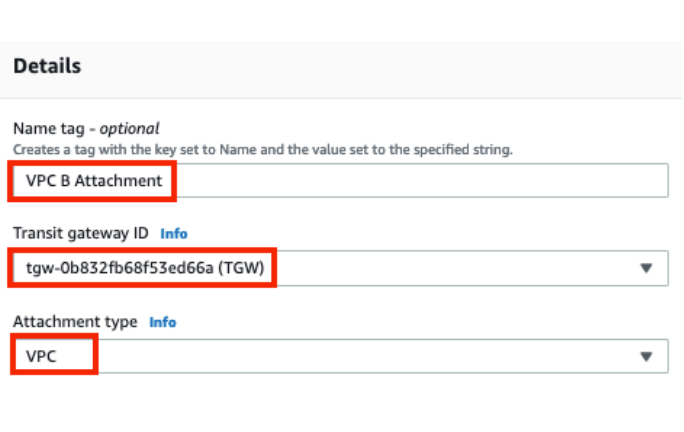
Transit gateway attachments (1)

Transit gateway attachment ID: tgw-attach-0e3fe3ad4d17f71c9

Name	Transit gateway attachment ID	Transit gateway ID	Resource type	Resource ID	State
VPC A Attachment	tgw-attach-0e3fe3ad4d17f71c9	tgw-0b832fb68f53ed66a	VPC	vpc-0a2f83322a1287eff	Pending

Create Transit Gateway Attachment for VPC B

1. Click **Create Transit Gateway Attachment**.
2. Enter `VPC B Attachment` as the **Name tag**.
3. Select the transit gateway from the dropdown for **Transit Gateway ID**.
4. Leave **Attachment Type** as `VPC`



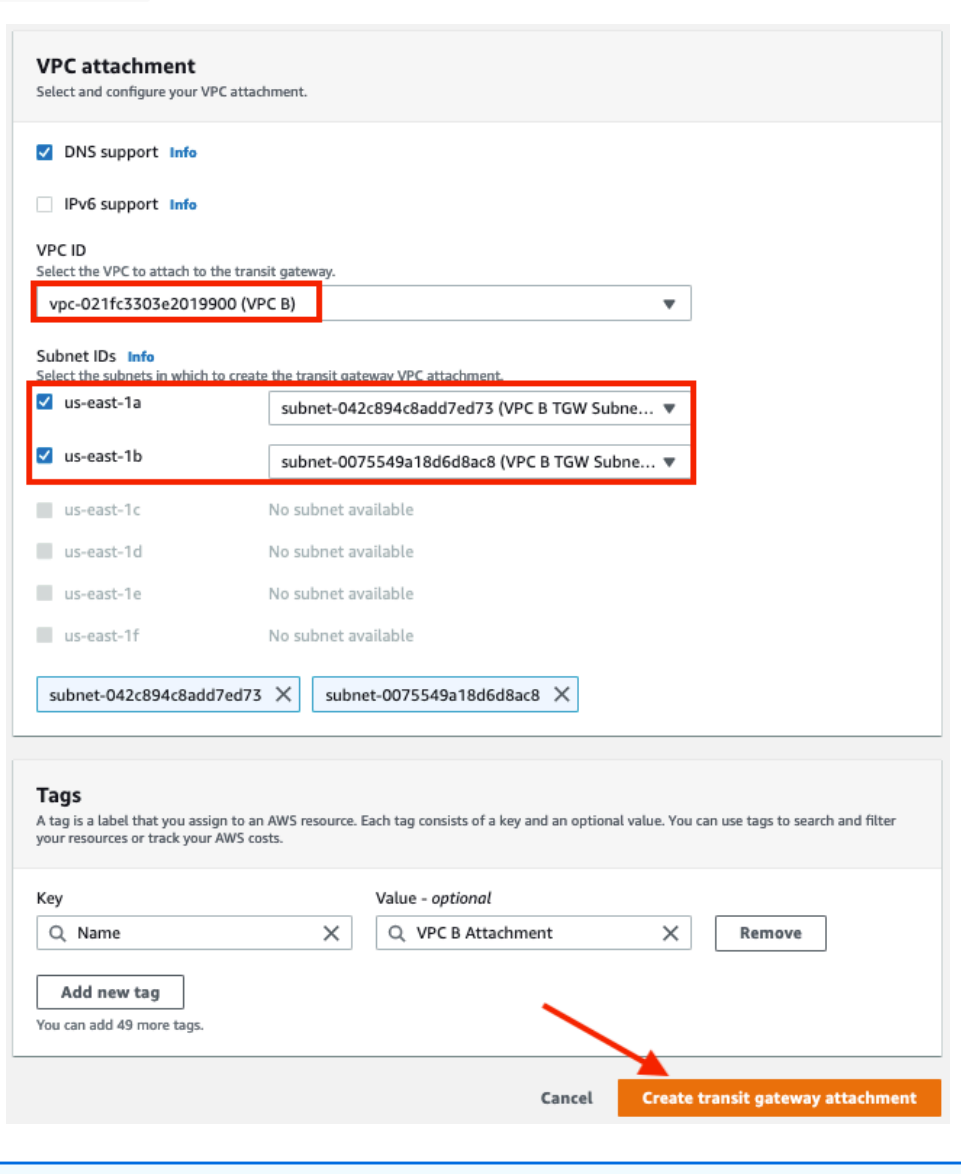
Details

Name tag - *optional*: VPC B Attachment

Transit gateway ID: tgw-0b832fb68f53ed66a (TGW)

Attachment type: VPC

5. Select **VPC B** from the **VPC ID** drop down
6. Select `VPC B TGW Subnet AZ1` and `VPC B TGW Subnet AZ2` for the **Subnet IDs**.



VPC attachment

Select and configure your VPC attachment.

☑ DNS support

☐ IPv6 support

VPC ID: vpc-021fc3303e2019900 (VPC B)

Subnet IDs

☑ us-east-1a subnet-042c894c8add7ed73 (VPC B TGW Subne…

☑ us-east-1b subnet-0075549a18d6d8ac8 (VPC B TGW Subne…

☐ us-east-1c No subnet available

☐ us-east-1d No subnet available

☐ us-east-1e No subnet available

☐ us-east-1f No subnet available

subnet-042c894c8add7ed73 subnet-0075549a18d6d8ac8

Tags

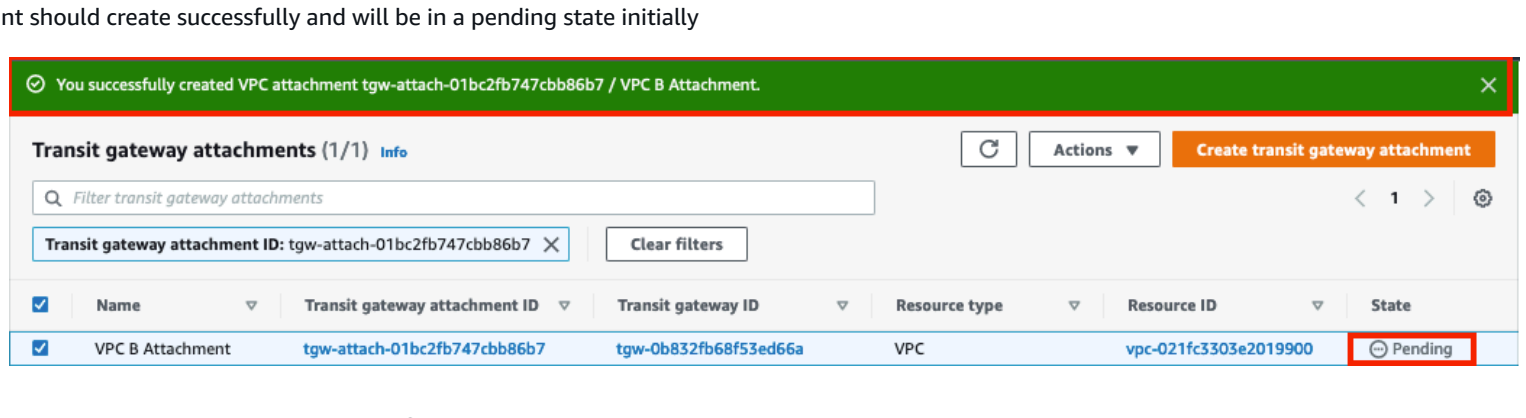
Key: Name Value - *optional*: VPC B Attachment

You can add 49 more tags.

Remove
Add new tag
Cancel
Create transit gateway attachment

> Note that the TGW subnets will not be selected by default, double check the subnets are the TGW ones.

7. Click **Create transit gateway attachment** on the bottom right corner.
8. VPC attachment should create successfully and will be in a pending state initially



You successfully created VPC attachment tgw-attach-01bc2fb747cbb86b7 / VPC B Attachment.

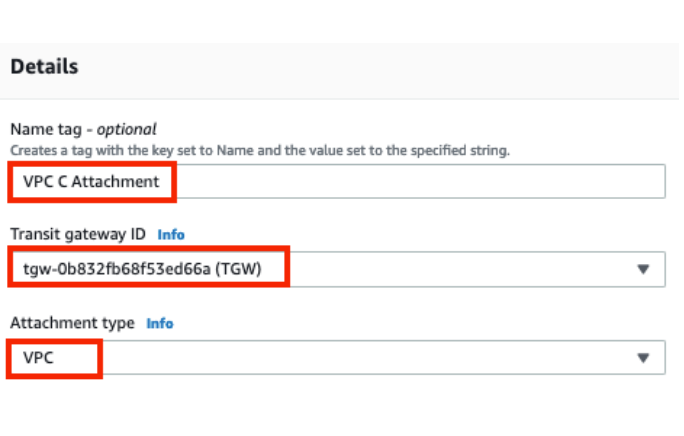
Transit gateway attachments (1/1)

Transit gateway attachment ID: tgw-attach-01bc2fb747cbb86b7

Name	Transit gateway attachment ID	Transit gateway ID	Resource type	Resource ID	State
VPC B Attachment	tgw-attach-01bc2fb747cbb86b7	tgw-0b832fb68f53ed66a	VPC	vpc-021fc3303e2019900	Pending

Create Transit Gateway Attachment for VPC C

1. Click **Create Transit Gateway Attachment**.
2. Name the attachment as `VPC C attachment`.
3. Select the transit gateway from the dropdown for **Transit Gateway ID**.
4. Leave **Attachment Type** as `VPC`



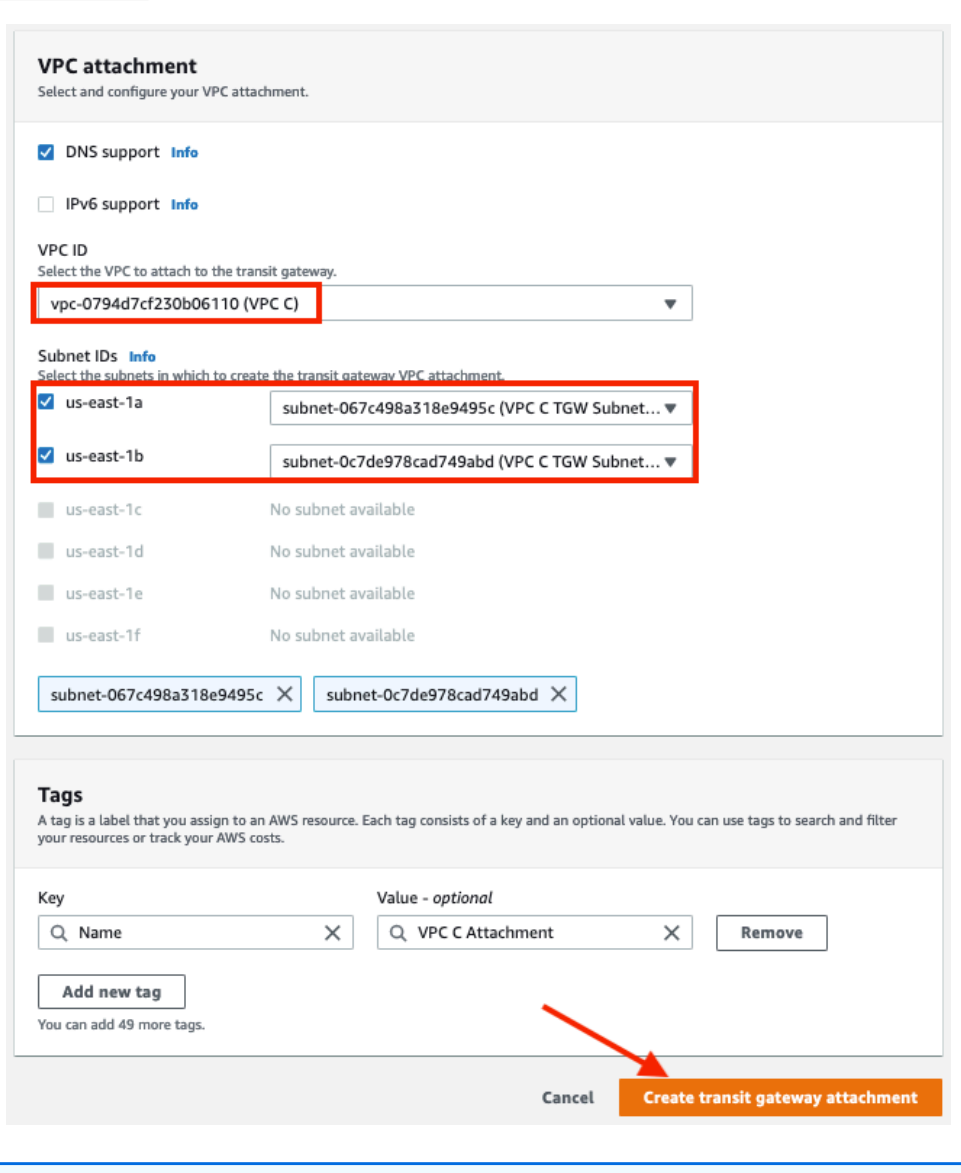
Details

Name tag - *optional*: VPC C Attachment

Transit gateway ID: tgw-0b832fb68f53ed66a (TGW)

Attachment type: VPC

5. Select `VPC C` from the **VPC ID** drop dowm.
6. Select `VPC C TGW Subnet AZ1` and `VPC C TGW Subnet AZ2` for the **Subnet IDs**.



VPC attachment

Select and configure your VPC attachment.

☑ DNS support

☐ IPv6 support

VPC ID: vpc-0794d7cf230b06110 (VPC C)

Subnet IDs

☑ us-east-1a subnet-067c498a318e9495c (VPC C TGW Subnet…

☑ us-east-1b subnet-0c7de978cad749abd (VPC C TGW Subnet…

☐ us-east-1c No subnet available

☐ us-east-1d No subnet available

☐ us-east-1e No subnet available

☐ us-east-1f No subnet available

subnet-067c498a318e9495c subnet-0c7de978cad749abd

Tags

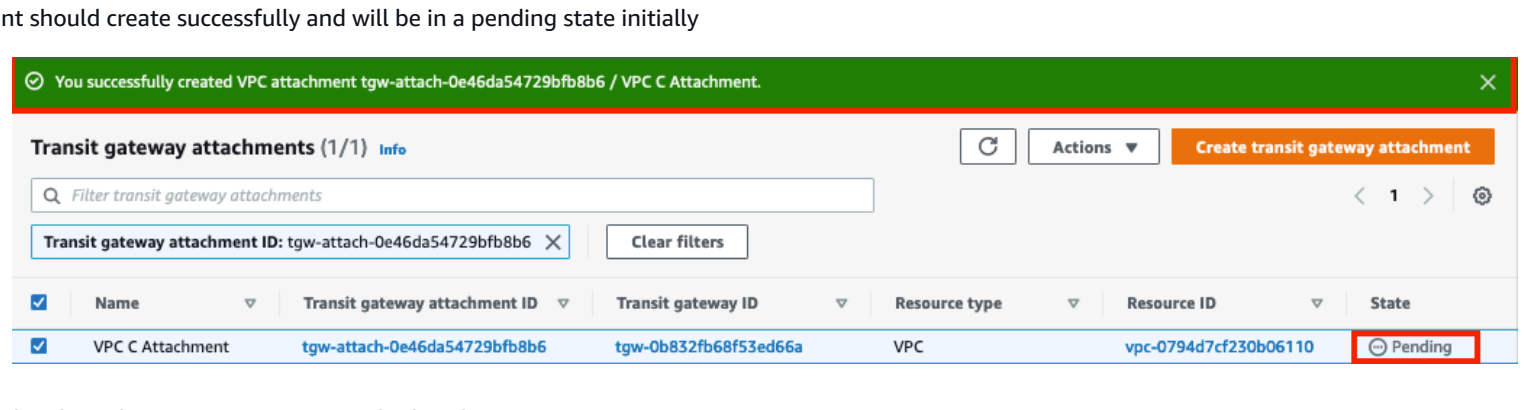
Key: Name Value - *optional*: VPC C Attachment

You can add 49 more tags.

Remove
Add new tag
Cancel
Create transit gateway attachment

> Note that the TGW subnets will not be selected by default, double check the subnets are the TGW ones.

7. Click **Create transit gateway attachment** on the bottom right corner.
8. VPC attachment should create successfully and will be in a pending state initially



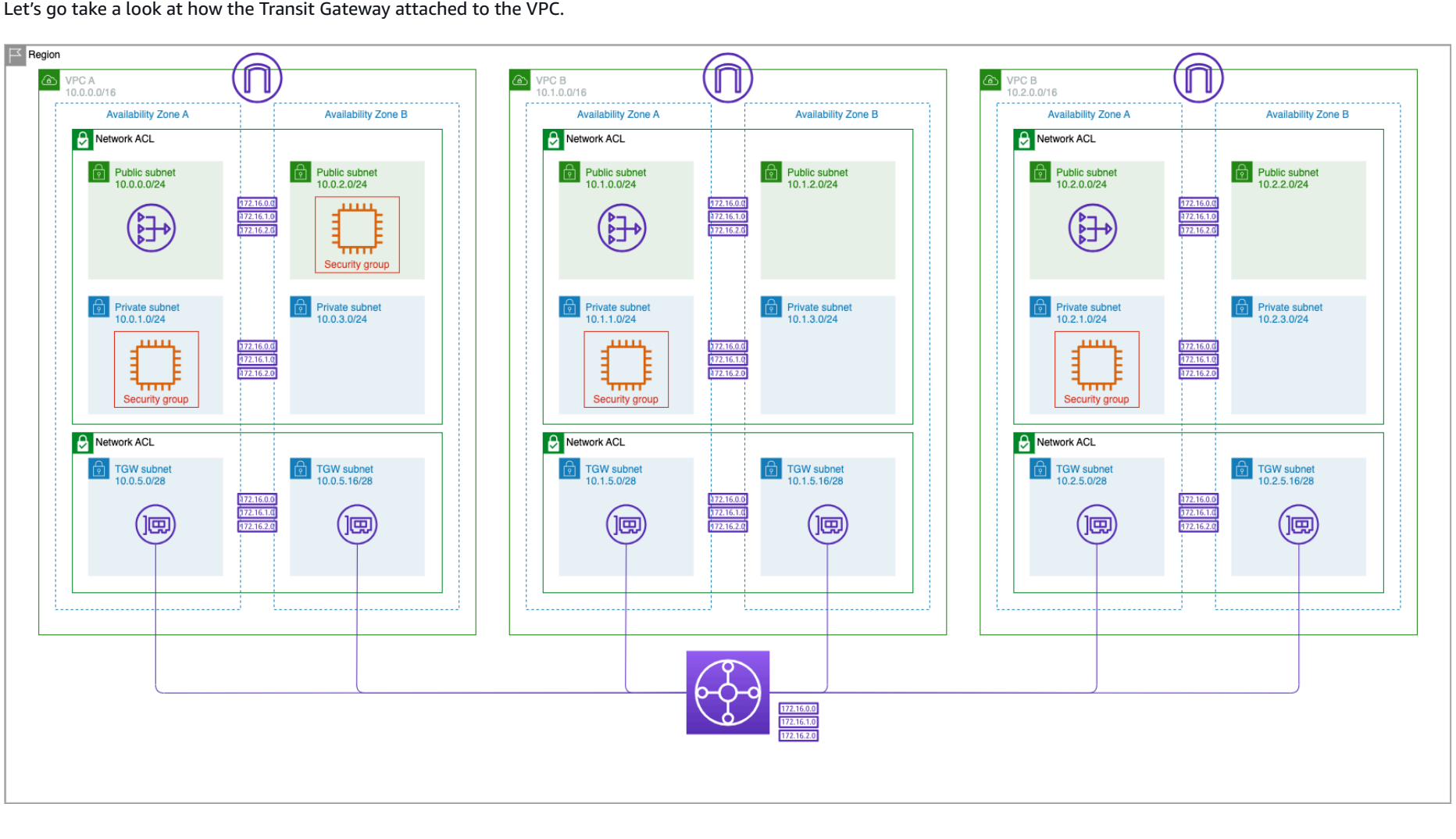
You successfully created VPC attachment tgw-attach-0e46da54729bfb8b6 / VPC C Attachment.

Transit gateway attachments (1/1)

Transit gateway attachment ID: tgw-attach-0e46da54729bfb8b6

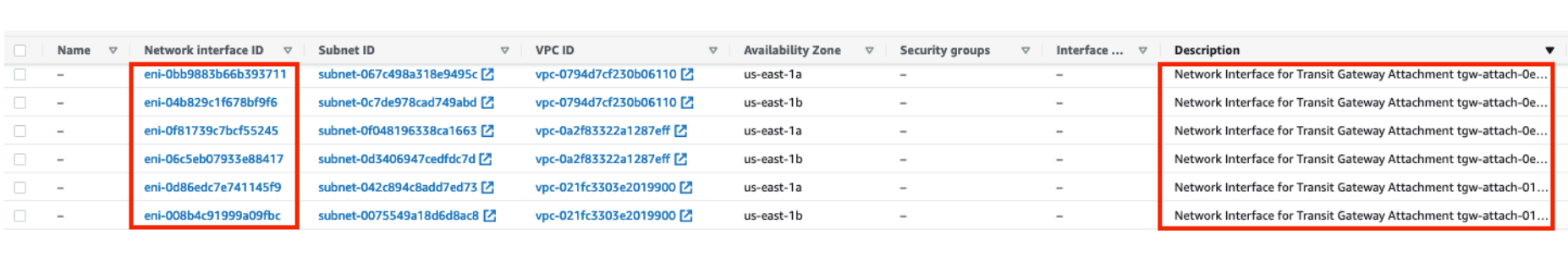
Name	Transit gateway attachment ID	Transit gateway ID	Resource type	Resource ID	State
VPC C Attachment	tgw-attach-0e46da54729bfb8b6	tgw-0b832fb68f53ed66a	VPC	vpc-0794d7cf230b06110	Pending

Let's go take a look at how the Transit Gateway attached to the VPC.



Region
VPC A 10.0.0.0/16
VPC B 10.1.0.0/16
VPC B 10.2.0.0/16
Availability Zone A
Availability Zone B
Network ACL
Public subnet 10.0.0.0/24
Public subnet 10.0.2.0/24
Private subnet 10.0.1.0/24
Private subnet 10.0.3.0/24
TGW subnet 10.0.5.0/28
TGW subnet 10.0.5.16/28
Public subnet 10.1.0.0/24
Public subnet 10.1.2.0/24
Private subnet 10.1.1.0/24
Private subnet 10.1.3.0/24
TGW subnet 10.1.5.0/28
TGW subnet 10.1.5.16/28
Public subnet 10.2.0.0/24
Public subnet 10.2.2.0/24
Private subnet 10.2.1.0/24
Private subnet 10.2.3.0/24
TGW subnet 10.2.5.0/28
TGW subnet 10.2.5.16/28
Security group

On the left in the EC2 Dashboard click on Network Interfaces



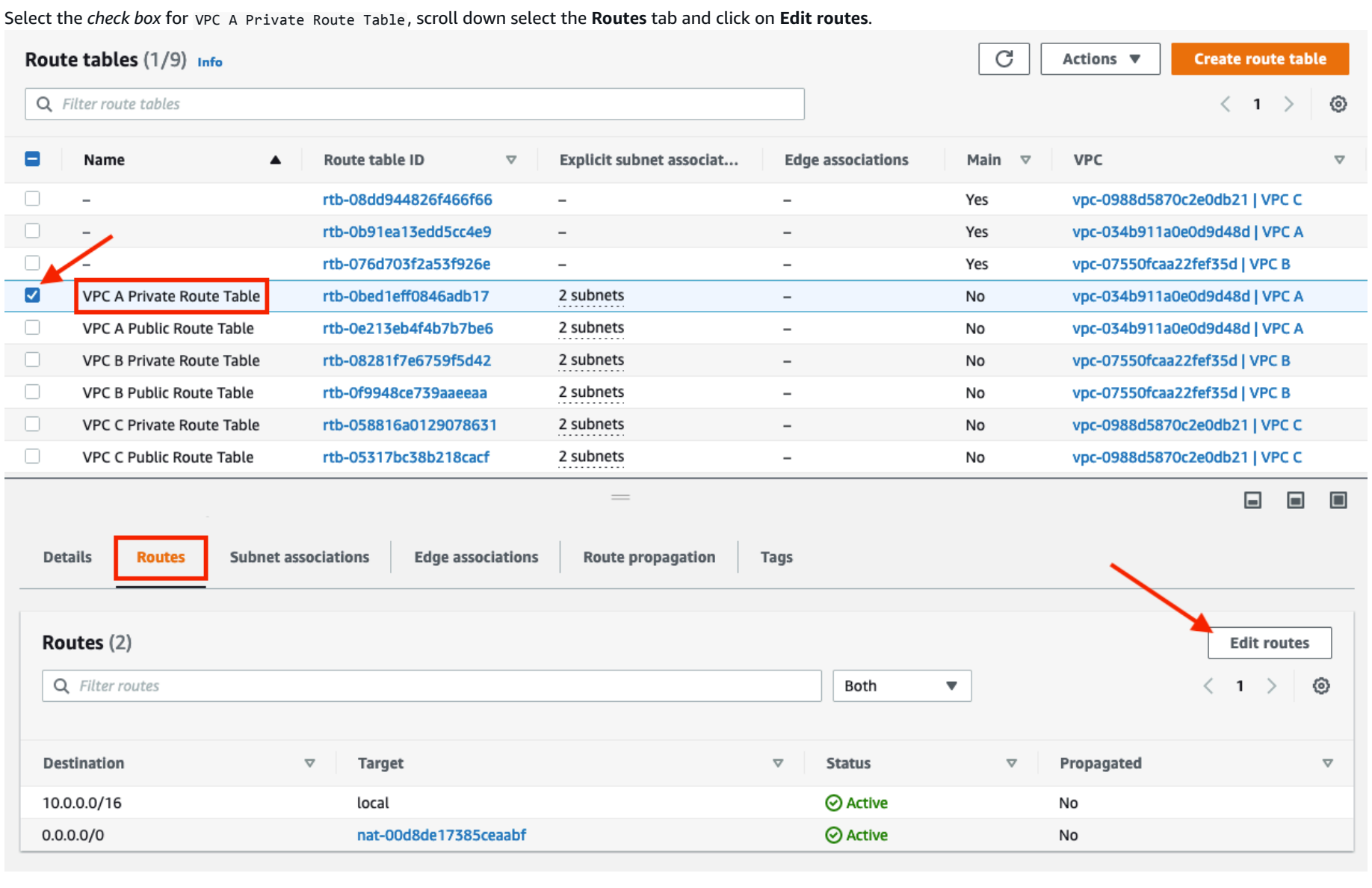
Name	Network interface ID	Subnet ID	VPC ID	Availability Zone	Security groups	Interface …	Description
–	eni-0bb9883b66b393711	subnet-067c498a318e9495c	vpc-0794d7cf230b06110	us-east-1a	–	–	Network Interface for Transit Gateway Attachment tgw-attach-0e…
–	eni-04b829c1f678bf9f6	subnet-0c7de978cad749abd	vpc-0794d7cf230b06110	us-east-1b	–	–	Network Interface for Transit Gateway Attachment tgw-attach-0e…
–	eni-0f81739c7bcf55245	subnet-0f048196338ca1663	vpc-0a2f83322a1287eff	us-east-1a	–	–	Network Interface for Transit Gateway Attachment tgw-attach-0e…
–	eni-06c5eb07933e88417	subnet-0d3406947cedfdc7d	vpc-0a2f83322a1287eff	us-east-1b	–	–	Network Interface for Transit Gateway Attachment tgw-attach-0e…
–	eni-0d86edc7e741145f9	subnet-042c894c8add7ed73	vpc-021fc3303e2019900	us-east-1a	–	–	Network Interface for Transit Gateway Attachment tgw-attach-01…
–	eni-008b4c91999a09fbc	subnet-0075549a18d6d8ac8	vpc-021fc3303e2019900	us-east-1b	–	–	Network Interface for Transit Gateway Attachment tgw-attach-01…

There are now six interfaces with a Description beginning `Network Interface for Transit Gateway Attachment...` representing the Elastic Network Interfaces that have been placed in each of the two Transit Gateway subnets in each of the three VPCs.

Now that we have attachments in all three VPCs, we need to add routes to their route tables to point traffic to the interfaces.

Add Routes to TGW in the VPC Route Tables

1. On the left in the VPC Dashboard click on Route Tables
2. Select the *check box* for `VPC A Private Route Table`, scroll down select the **Routes** tab and click on **Edit routes**.



Route tables (1/9)

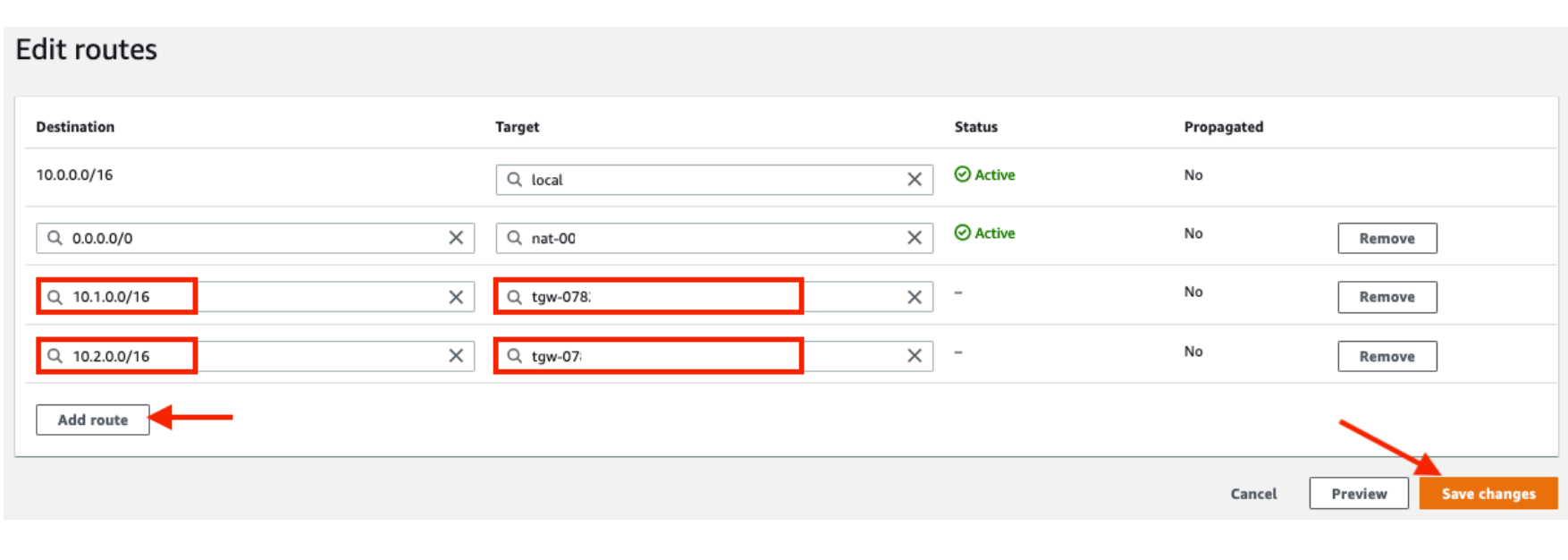
	Name	Route table ID	Explicit subnet associat…	Edge associations	Main	VPC
☐	–	rtb-08dd944826f466f66	–	–	Yes	vpc-0988d5870c2e0db21 | VPC C
☐	–	rtb-0b91ea13edd5cc4e9	–	–	Yes	vpc-034b911a0e0d9d48d | VPC A
☐	–	rtb-076d703f2a53f926e	–	–	Yes	vpc-07550fcaa22fef35d | VPC B
☑	VPC A Private Route Table	rtb-0bed1eff0846adb17	2 subnets	–	No	vpc-034b911a0e0d9d48d | VPC A
☐	VPC A Public Route Table	rtb-0e213eb4f4b7b7be6	2 subnets	–	No	vpc-034b911a0e0d9d48d | VPC A
☐	VPC B Private Route Table	rtb-08281f7e6759f5d42	2 subnets	–	No	vpc-07550fcaa22fef35d | VPC B
☐	VPC B Public Route Table	rtb-0f9948ce739aaeeaa	2 subnets	–	No	vpc-07550fcaa22fef35d | VPC B
☐	VPC C Private Route Table	rtb-058816a0129078631	2 subnets	–	No	vpc-0988d5870c2e0db21 | VPC C
☐	VPC C Public Route Table	rtb-05317bc38b218cacf	2 subnets	–	No	vpc-0988d5870c2e0db21 | VPC C

Actions
Create route table
Details
Routes
Subnet associations
Edge associations
Route propagation
Tags
Edit routes
Both

Routes (2)

Destination	Target	Status	Propagated
10.0.0.0/16	local	Active	No
0.0.0.0/0	nat-00d8de17385ceaabf	Active	No

3. Add a route to VPC B in the **VPC A Private Route Table** using a Destination of `10.1.0.0/16` with a Target as the transit gateway.
4. Add a route to VPC C in the **VPC A Private Route Table** using a Destination of `10.2.0.0/16` with a Target as the transit gateway.
5. Click **Save routes**.

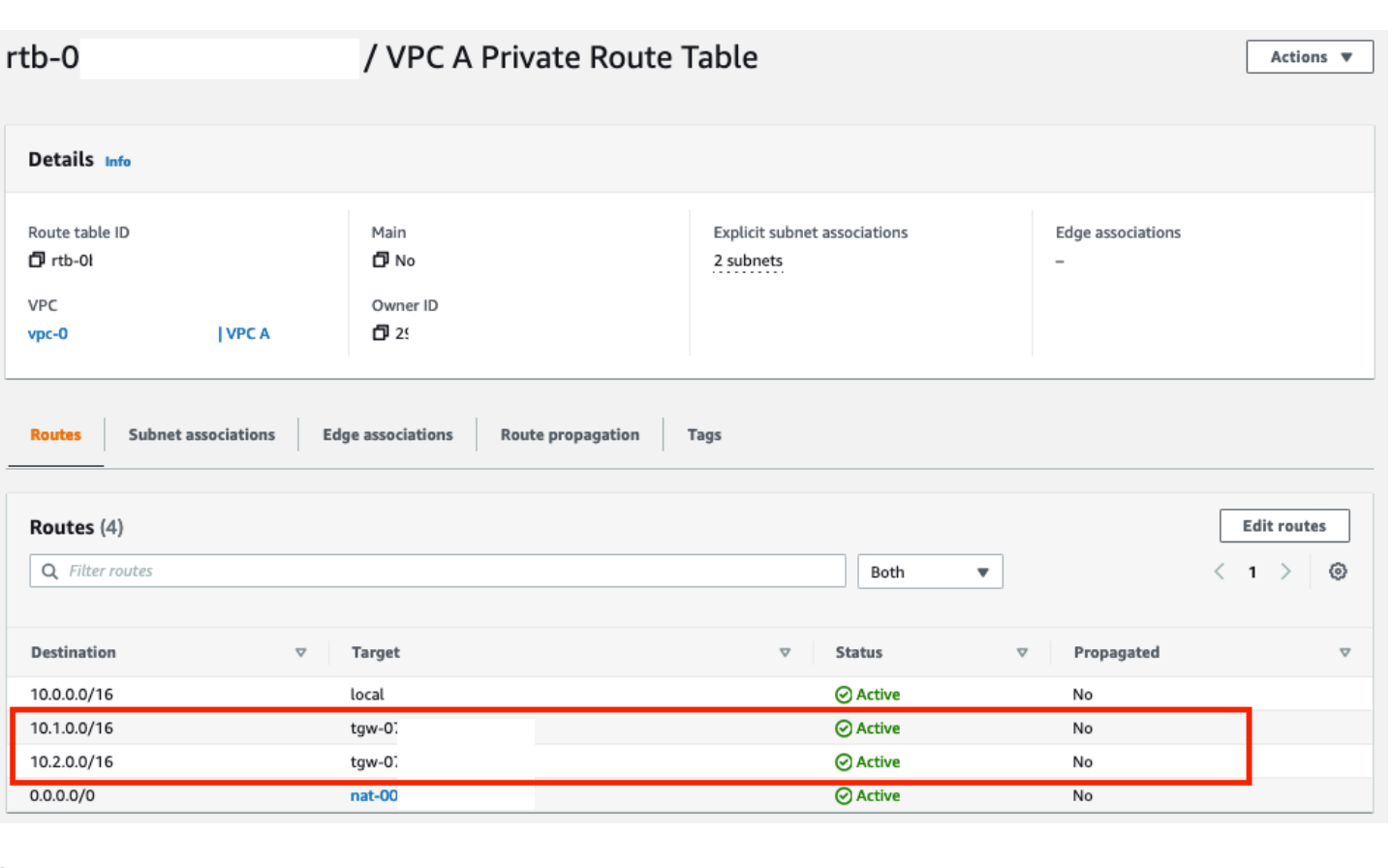


Edit routes

Destination	Target	Status	Propagated
10.0.0.0/16	local	Active	No
0.0.0.0/0	nat-00	Active	No
10.1.0.0/16	tgw-078	–	No
10.2.0.0/16	tgw-07	–	No

Remove
Add route
Cancel
Preview
Save changes

6. Confirm that the routes have been added to the route table.



rtb-0[redacted] / VPC A Private Route Table

Details

Route table ID: rtb-0I

Main: No

Explicit subnet associations: 2 subnets

Edge associations: –

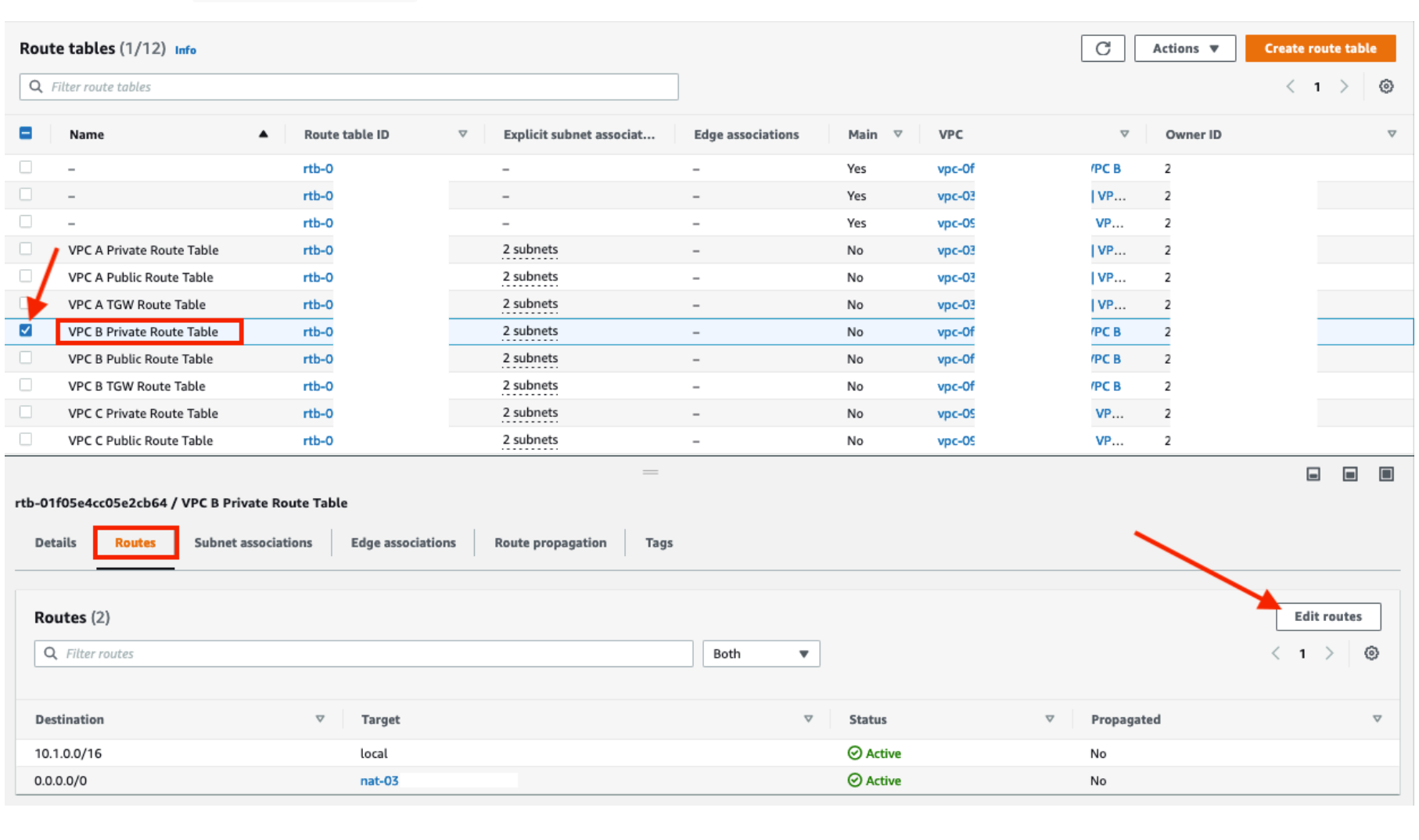
VPC: vpc-0 | VPC A

Owner ID: 2!

Routes (4)

Destination	Target	Status	Propagated
10.0.0.0/16	local	Active	No
10.1.0.0/16	tgw-0	Active	No
10.2.0.0/16	tgw-0	Active	No
0.0.0.0/0	nat-00	Active	No

7. Click on **Route tables**
8. Select the *check box* for `VPC B Private Route Table`, scroll down select the **Routes** tab and click on **Edit routes**.



Route tables (1/12)

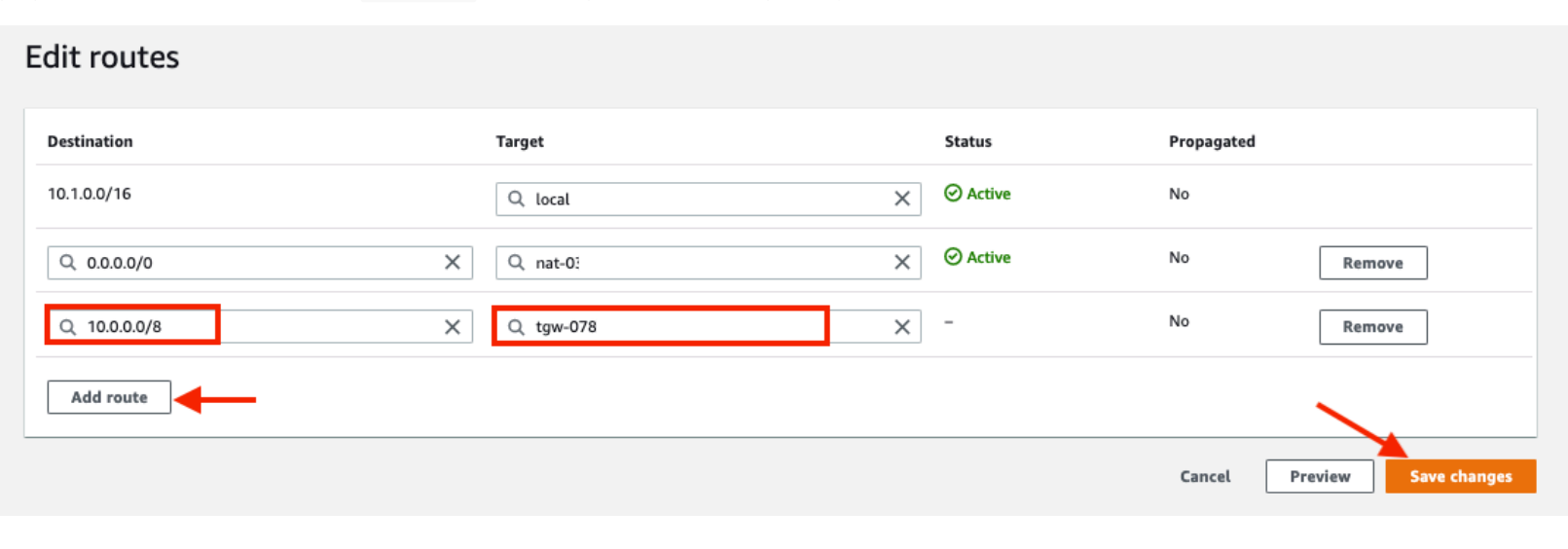
	Name	Route table ID	Explicit subnet associat…	Edge associations	Main	VPC	Owner ID
☐	–	rtb-0	–	–	Yes	vpc-0f | VPC B	2
☐	–	rtb-0	–	–	Yes	vpc-03 | VP…	2
☐	–	rtb-0	–	–	Yes	vpc-05 | VP…	2
☐	VPC A Private Route Table	rtb-0	2 subnets	–	No	vpc-03 | VP…	2
☐	VPC A Public Route Table	rtb-0	2 subnets	–	No	vpc-03 | VP…	2
☐	VPC A TGW Route Table	rtb-0	2 subnets	–	No	vpc-03 | VP…	2
☑	VPC B Private Route Table	rtb-0	2 subnets	–	No	vpc-0f | VPC B	2
☐	VPC B Public Route Table	rtb-0	2 subnets	–	No	vpc-0f | VPC B	2
☐	VPC B TGW Route Table	rtb-0	2 subnets	–	No	vpc-0f | VPC B	2
☐	VPC C Private Route Table	rtb-0	2 subnets	–	No	vpc-05 | VP…	2
☐	VPC C Public Route Table	rtb-0	2 subnets	–	No	vpc-05 | VP…	2

rtb-01f05e4cc05e2cb64 / VPC B Private Route Table

Routes (2)

Destination	Target	Status	Propagated
10.1.0.0/16	local	Active	No
0.0.0.0/0	nat-03	Active	No

9. Add an aggregate route with a Destination of `10.0.0.0/8` with a Target as the transit gateway. Click **Save routes**.

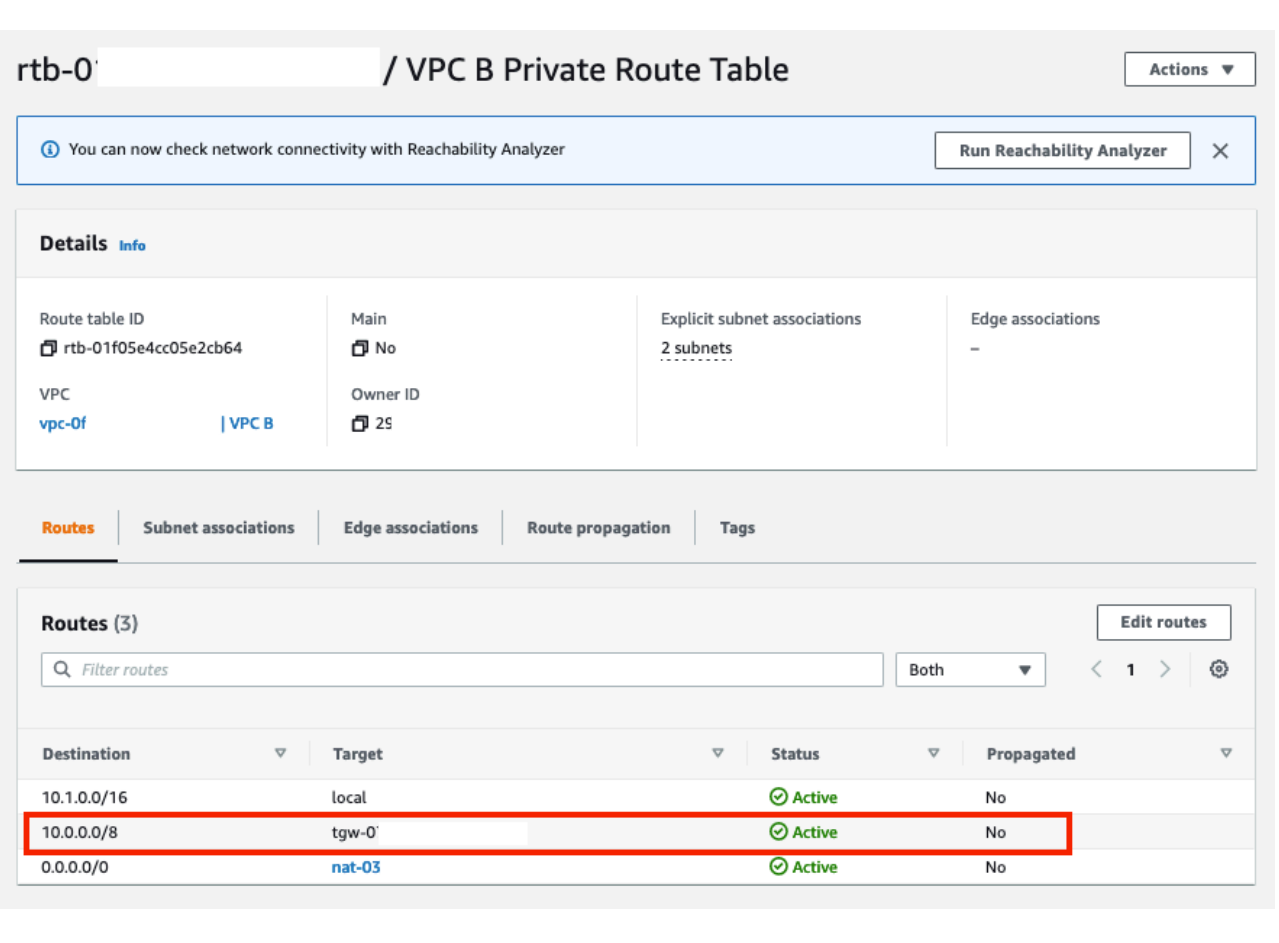


Edit routes

Destination	Target	Status	Propagated
10.1.0.0/16	local	Active	No
0.0.0.0/0	nat-0:	Active	No
10.0.0.0/8	tgw-078	–	No

Remove
Add route
Cancel
Preview
Save changes

10. Confirm that the route has been added to the **VPC B Private Route Table**.



rtb-0[redacted] / VPC B Private Route Table

You can now check network connectivity with Reachability Analyzer

Details

Route table ID: rtb-01f05e4cc05e2cb64

Main: No

Explicit subnet associations: 2 subnets

Edge associations: –

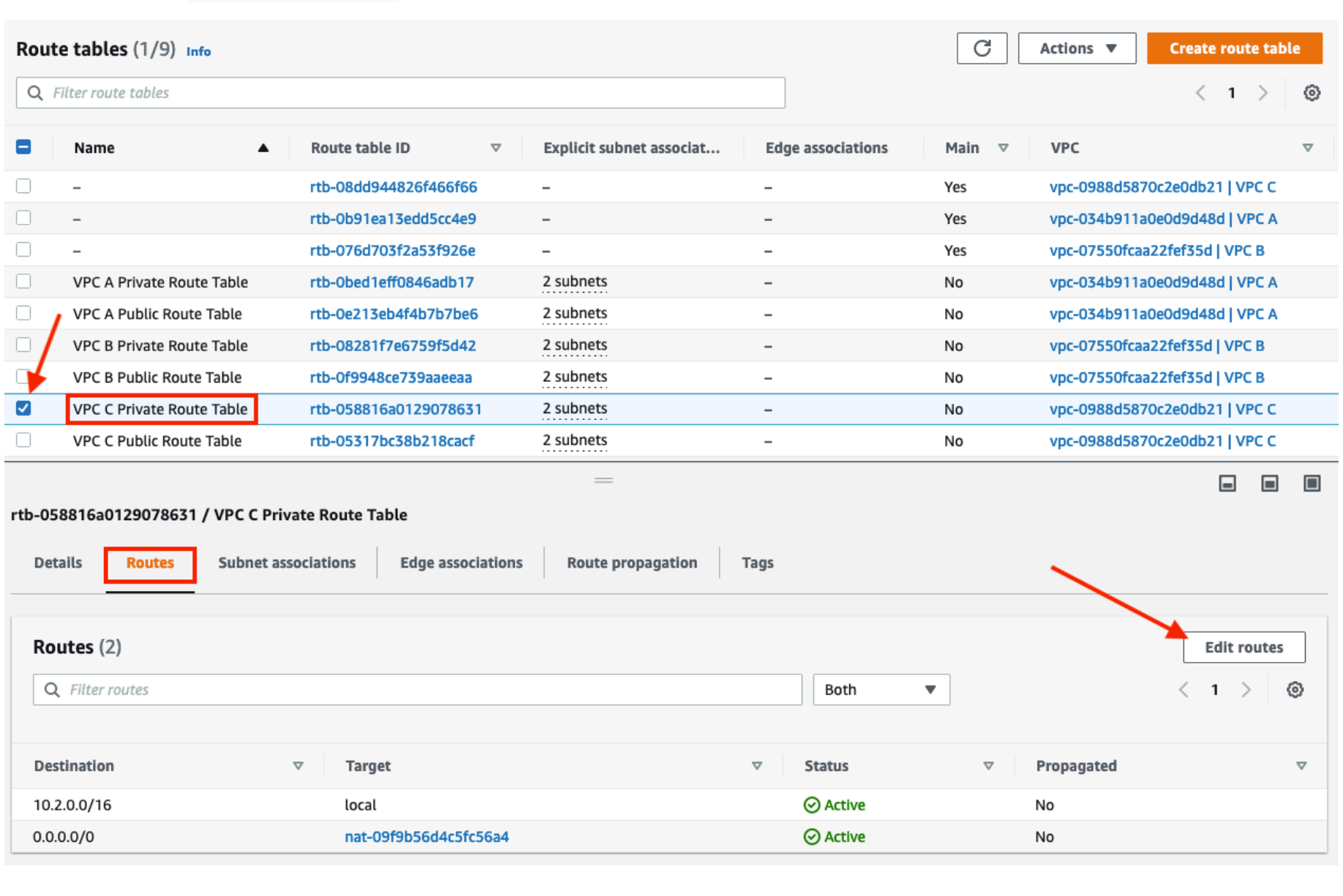
VPC: vpc-0f | VPC B

Owner ID: 25

Routes (3)

Destination	Target	Status	Propagated
10.1.0.0/16	local	Active	No
10.0.0.0/8	tgw-0	Active	No
0.0.0.0/0	nat-03	Active	No

11. Click on **Route tables**
12. Select the *check box* for `VPC C Private Route Table`, scroll down select the **Routes** tab and click on **Edit routes**.



Route tables (1/9)

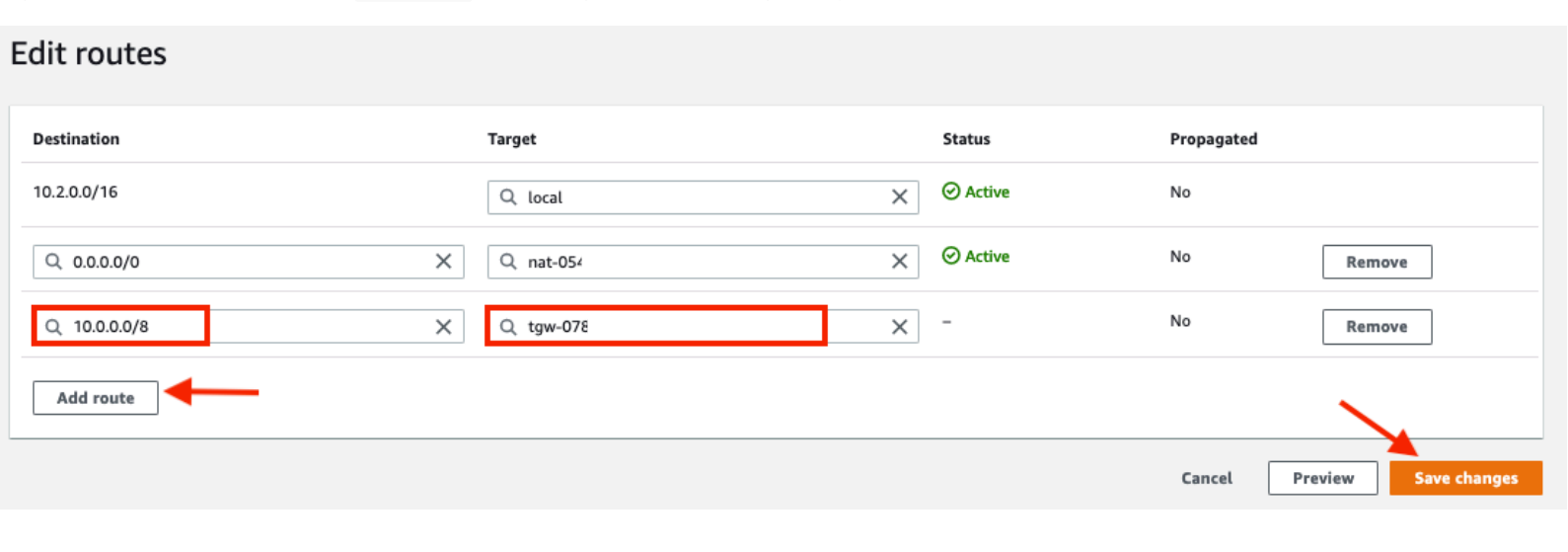
	Name	Route table ID	Explicit subnet associat…	Edge associations	Main	VPC
☐	–	rtb-08dd944826f466f66	–	–	Yes	vpc-0988d5870c2e0db21 | VPC C
☐	–	rtb-0b91ea13edd5cc4e9	–	–	Yes	vpc-034b911a0e0d9d48d | VPC A
☐	–	rtb-076d703f2a53f926e	–	–	Yes	vpc-07550fcaa22fef35d | VPC B
☐	VPC A Private Route Table	rtb-0bed1eff0846adb17	2 subnets	–	No	vpc-034b911a0e0d9d48d | VPC A
☐	VPC A Public Route Table	rtb-0e213eb4f4b7b7be6	2 subnets	–	No	vpc-034b911a0e0d9d48d | VPC A
☐	VPC B Private Route Table	rtb-08281f7e6759f5d42	2 subnets	–	No	vpc-07550fcaa22fef35d | VPC B
☐	VPC B Public Route Table	rtb-0f9948ce739aaeeaa	2 subnets	–	No	vpc-07550fcaa22fef35d | VPC B
☑	VPC C Private Route Table	rtb-058816a0129078631	2 subnets	–	No	vpc-0988d5870c2e0db21 | VPC C
☐	VPC C Public Route Table	rtb-05317bc38b218cacf	2 subnets	–	No	vpc-0988d5870c2e0db21 | VPC C

rtb-058816a0129078631 / VPC C Private Route Table

Routes (2)

Destination	Target	Status	Propagated
10.2.0.0/16	local	Active	No
0.0.0.0/0	nat-09f9b56d4c5fc56a4	Active	No

13. Add an aggregate route with a Destination of `10.0.0.0/8` with a Target as the transit gateway. Click Save routes.

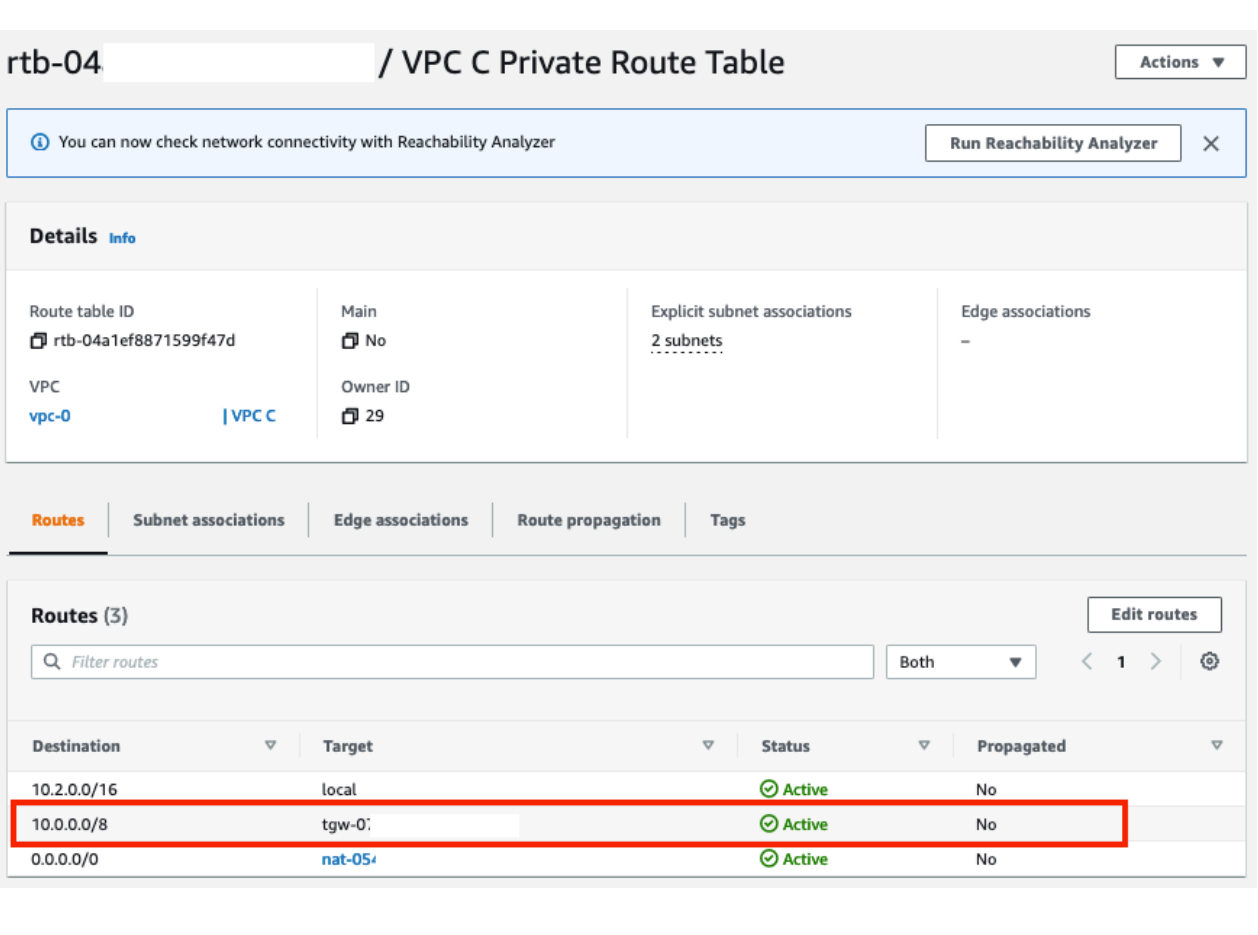


Edit routes

Destination	Target	Status	Propagated
10.2.0.0/16	local	Active	No
0.0.0.0/0	nat-05	Active	No
10.0.0.0/8	tgw-078	–	No

Remove
Add route
Cancel
Preview
Save changes

14. Confirm that the route has been added to the **VPC C Private Route Table**.



rtb-04[redacted] / VPC C Private Route Table

You can now check network connectivity with Reachability Analyzer

Details

Route table ID: rtb-04a1ef8871599f47d

Main: No

Explicit subnet associations: 2 subnets

Edge associations: –

VPC: vpc-0 | VPC C

Owner ID: 29

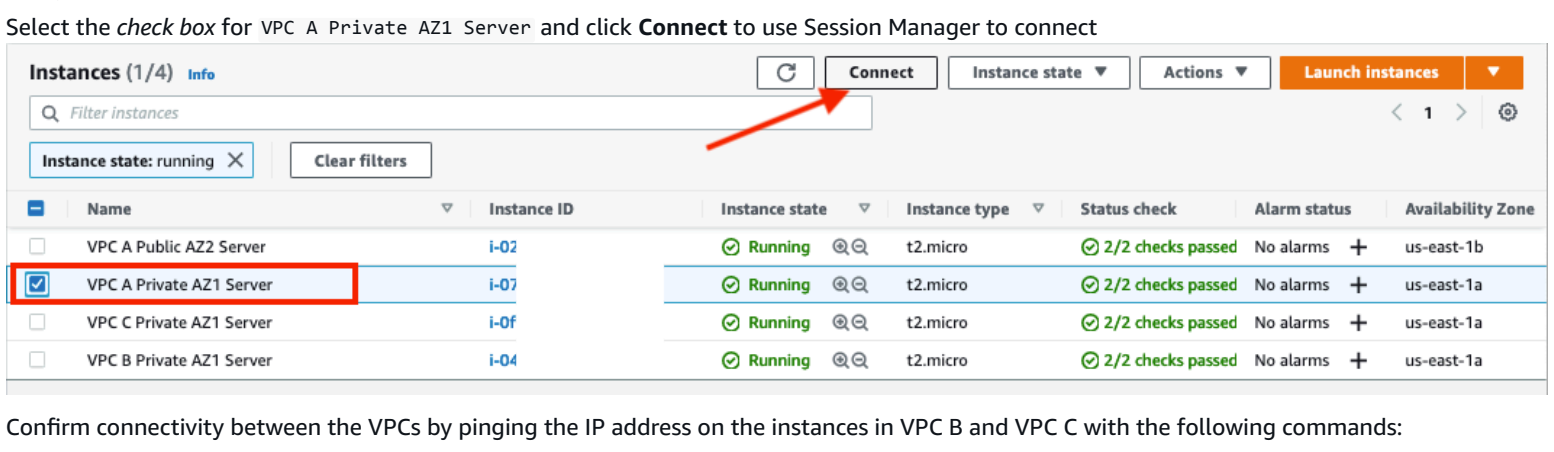
Routes (3)

Destination	Target	Status	Propagated
10.2.0.0/16	local	Active	No
10.0.0.0/8	tgw-0	Active	No
0.0.0.0/0	nat-05	Active	No

Test Connectivity

Now let's test the connectivity between the instances in the private subnets in VPC A, B & C.

1. Navigate to Instances in the EC2 Dashboard
2. Select the *check box* for `VPC A Private AZ1 Server` and click **Connect** to use Session Manager to connect



Instances (1/4)

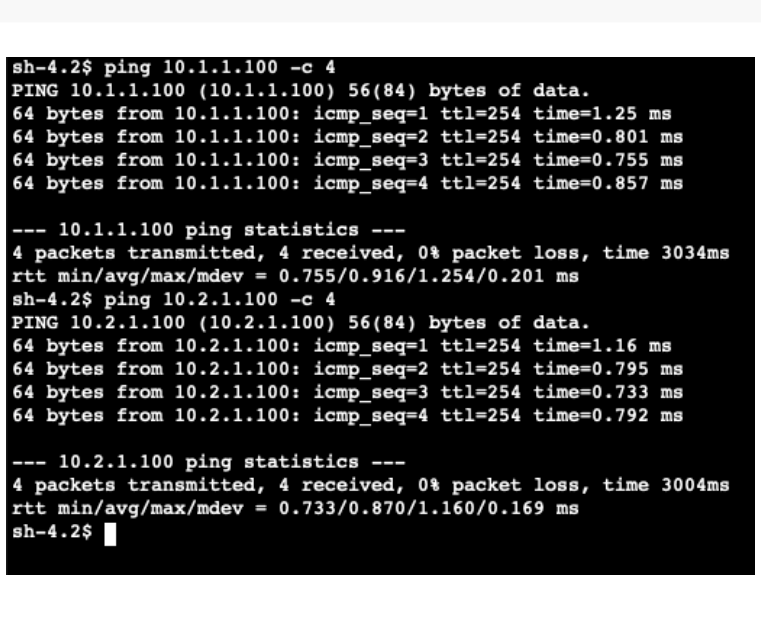
Instance state: running

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
☐	VPC A Public AZ2 Server	i-02	Running	t2.micro	2/2 checks passed	No alarms	us-east-1b
☑	VPC A Private AZ1 Server	i-07	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a
☐	VPC C Private AZ1 Server	i-0f	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a
☐	VPC B Private AZ1 Server	i-04	Running	t2.micro	2/2 checks passed	No alarms	us-east-1a

Connect
Instance state
Actions
Launch instances
Clear filters

3. Confirm connectivity between the VPCs by pinging the IP address on the instances in VPC B and VPC C with the following commands:

```
ping 10.1.1.100 -c 5
ping 10.2.1.100 -c 5
```



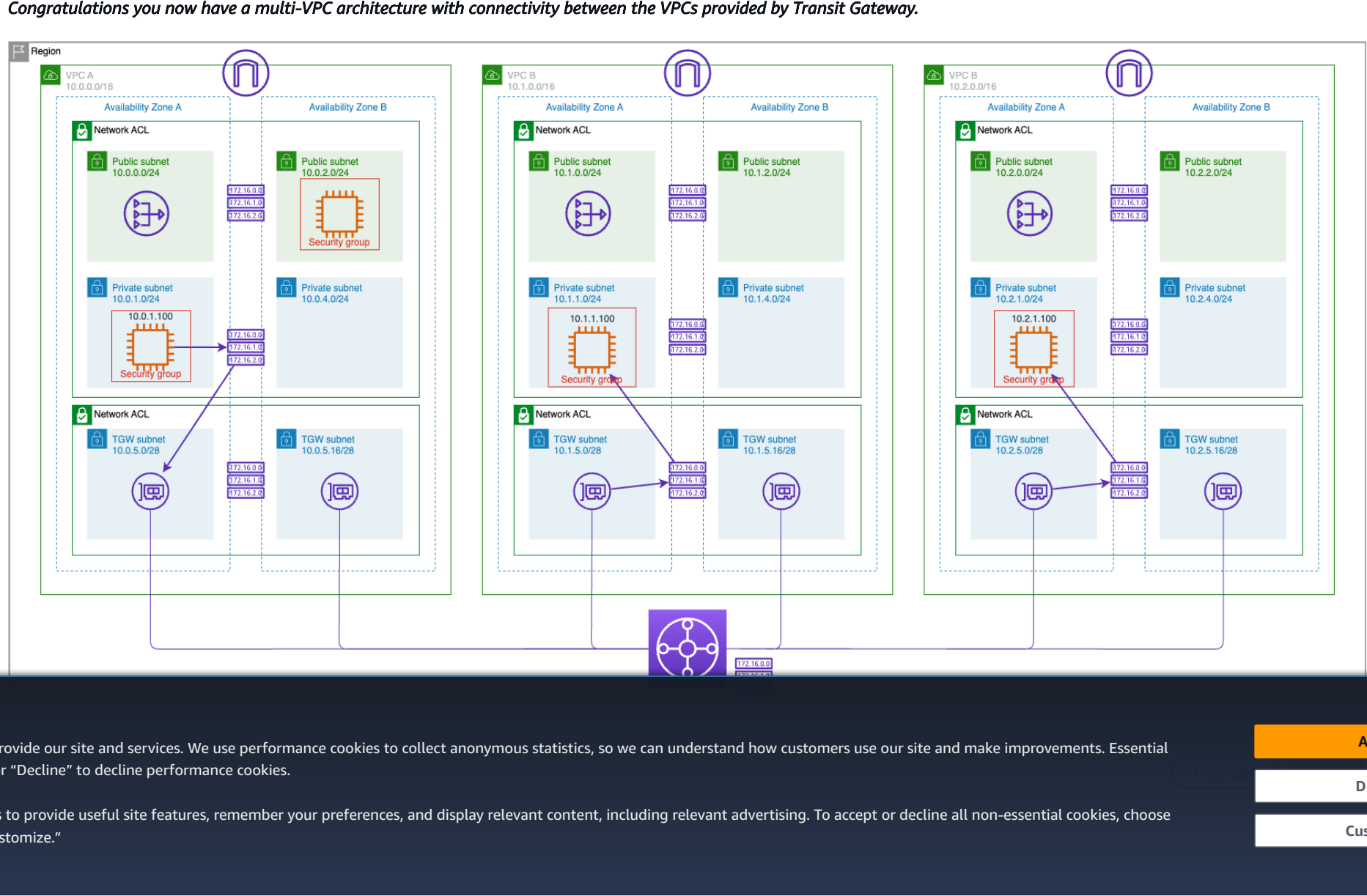
```
sh-4.2$ ping 10.1.1.100 -c 4
PING 10.1.1.100 (10.1.1.100) 56(84) bytes of data.
64 bytes from 10.1.1.100: icmp_seq=1 ttl=254 time=1.25 ms
64 bytes from 10.1.1.100: icmp_seq=2 ttl=254 time=0.801 ms
64 bytes from 10.1.1.100: icmp_seq=3 ttl=254 time=0.755 ms
64 bytes from 10.1.1.100: icmp_seq=4 ttl=254 time=0.857 ms

--- 10.1.1.100 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3034ms
rtt min/avg/max/mdev = 0.755/0.916/1.254/0.201 ms
sh-4.2$ ping 10.2.1.100 -c 4
PING 10.2.1.100 (10.2.1.100) 56(84) bytes of data.
64 bytes from 10.2.1.100: icmp_seq=1 ttl=254 time=1.16 ms
64 bytes from 10.2.1.100: icmp_seq=2 ttl=254 time=0.795 ms
64 bytes from 10.2.1.100: icmp_seq=3 ttl=254 time=0.733 ms
64 bytes from 10.2.1.100: icmp_seq=4 ttl=254 time=0.792 ms

--- 10.2.1.100 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 0.733/0.870/1.160/0.169 ms
sh-4.2$ 
```

4. You should see a response from both EC2 instances

Congratulations you now have a multi-VPC architecture with connectivity between the VPCs provided by Transit Gateway.



Region
VPC A 10.0.0.0/16
VPC B 10.1.0.0/16
VPC B 10.2.0.0/16
Availability Zone A
Availability Zone B
Network ACL
Public subnet 10.0.0.0/24
Public subnet 10.0.2.0/24
Private subnet 10.0.1.0/24
Private subnet 10.0.4.0/24
10.0.1.100
TGW subnet 10.0.5.0/28
TGW subnet 10.0.5.16/28
Public subnet 10.1.0.0/24
Public subnet 10.1.2.0/24
Private subnet 10.1.1.0/24
Private subnet 10.1.4.0/24
10.1.1.100
TGW subnet 10.1.5.0/28
TGW subnet 10.1.5.16/28
Public subnet 10.2.0.0/24
Public subnet 10.2.2.0/24
Private subnet 10.2.1.0/24
Private subnet 10.2.4.0/24
10.2.1.100
TGW subnet 10.2.5.0/28
TGW subnet 10.2.5.16/28
Security group

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